Does 0DTE Options Trading Increase Volatility?

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ABSTRACT

This paper examines the impact of Zero-Day-to-Expiration (0DTE) options trading on stock market volatility. The monthly trading volume of 0DTE options linked to indices increased from .08 million contracts in January 2011 to 34.4 million contracts in August 2023 and now accounts for 48% of the trading in index options. Using the staggered introduction of index weekly options as an instrument variable, we show that a one standard deviation increase in 0DTE options trading leads to 9.10% increase relative to the mean value of volatility, which is 15.91% of its standard deviation. Even after controlling for option market makers' gamma hedging, we find that the overall impact on volatility remains positive and is primarily driven by speculative retail investors.

Keywords: Zero-Day-to-Expiration (0DTE) Options, Volatility, Market Efficiency

JEL classification: G12, G13, G14, G17

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1. Introduction

In December 2010, the Chicago Board Options Exchange (CBOE) introduced short-dated option products, SPXW, linked to the S&P 500 index. Since then, short-dated option products have been introduced for other major indices after 2020, including the NASDAQ-100 and Russell 2000 index. These new option products have a life ranging from one week to one month, dramatically shorter than the previously available index-linked option, whose lifespans range from three months to three years. The increase in options with varying expiration days has led to more opportunities for investors to trade near the expiration of an option, especially Zero-Day-to-Expiration (0DTE) options. 0DTE options are referred to as options traded on the last day before expiring. 0DTE options trading has recently exploded in popularity. While there is extant literature exploring the influence of options on their underlying assets[1], there is little work examining the impact of short-term expiration options on the underlying assets. This paper examines whether 0DTE options trading affects the volatility of the stock market.

There is a long-running debate about whether derivative securities impact their underlying asset. The traditional view is that the underlying assets are independent of derivative securities that they are written on. For example, Black and Sholes (1973) framework argues that since all derivative securities can be spanned by underlying assets, they are redundant products that do not affect their underlying assets. Furthermore, Cox and Ross (1976) support this idea by modeling the stochastic price process independently and showing that the price of an option is a function of its underlying stock. The model implies that option

[1] The general relationship between options trading and the volatility of the underlying asset is well studied (Day and Lewis 1988; Skinner 1989; Conrad 1989; Harris 1989; Kamara, Miller, and Siegel 1992; Kumar et al. 1998; Ni, Pan, and Poteshman 2008; Muravyev, Pearson, and Broussard 2013). Regarding the relationship between the volatility of underlying assets and options trading, the literature reveals mixed and divergent results. For instance, Kumar et al. (1998), Skinner (1989), and Conrad (1989) argue that options trading leads to the reduction in the volatility of underlying assets. On the other hand, Day and Lewis (1988), Harris (1989), and Ni et al. (2008) document that options trading is positively related to the volatility of underlying assets. However, Kamara et al. (1992), Long et al. (1994), Darrat and Rahman (1995), and Muravyev et al. (2013) find no clear evidence on the relationship.

contracts do not affect the price of the underlying stock because an underlying stock and an option written on it are perfectly correlated.

In contrast, the alternative view argues that derivative securities affect their underlying asset. Danthine (1978) provides a theoretical model showing that the derivative markets contribute to the reduction in spot volatility since the reduced transaction costs in derivatives trading allow the informed traders to respond to the mispricing more efficiently. Stein (1987) argues that the introduction of a future market decreases spot price volatility because of risk sharing between speculators and informed traders. He concludes that the introduction would lead to heightened volatility when the speculators trade based on information with noise, but hedgers respond to this noise in the speculative trades. Ross (1989) models information flowing from the derivatives market and affecting the underlying market. If information is contained in the derivatives trading, then the price of the spot market is affected, and the volatility of the spot market increases. More recently, Bhamra and Uppal (2009) provide a theoretical framework whereby the introduction of a non-redundant derivative causes an increase in the volatility of the spot market.

While theories have mixed results, we argue that 0DTE options trading increases market volatility based on empirical evidence and theoretical literature on retail investor participation. Given that retail investors constitute the majority of options trading, and they are commonly regarded as uninformed or noise traders in financial markets, their trading activity likely injects noise into security prices, causing deviations from their fundamental value. This noise, in turn, contributes to increased volatility. In other words, we conjecture that 0DTE options are a conduit for noise trading, therefore increasing market volatility.

It is documented that retail investors are the dominant participants in 0DTE options trading. Most recently, CBOE Global Markets reports that retail investors account for approximately 50–60% of SPX 0DTE trading volume.[2] Consistent with this, Bryzgalova et al. (2023) find that retail investors are now the largest

[2] https://www.cboe.com/insights/posts/0-dt-es-decoded-positioning-trends-and-market-impact

investors in options trading, and their proportion exceeds 60% of the total options trading between November 2019 and June 2021. Similarly, a 2023 report by the New York Stock Exchange documents that retail investors account for 45% of the options market, and their trading is more pronounced in short-term expiration options. Specifically, retail traders make up 51% of 0DTE options trading volume, whereas their share in options with one to three months to expiration accounts for 34%.[3]

Retail traders are typically considered noise traders who are speculative and uninformed (Lakonishok et al. 2007; Liu et al. 2020). These characteristics are attributed to the distortion of stock prices and the heightened level of price volatility (Shleifer and Summers 1990; Shleifer and Vishny 1997; Kumar and Lee 2006). For example, De Long et al. (1990) argue that the unpredictability of noise traders' beliefs can generate risk in the prices and prevent rational arbitrageurs from actively trading against them, leading to an extremely high volatility of stock prices. Barberis, Shleifer, and Wurgler (2005) provide a theoretical framework where different classes of assets can display co-movement because their prices deviate from their fundamental values under the existence of noise traders. Kumar and Lee (2006) reinforce the argument of Barberis, Shleifer, and Wurgler (2005) by providing empirical evidence that retail investors' sentiment can influence stock prices, resulting in a deviation from their fundamental values. In line with this idea, more recently, Bloomfield, O'Hara, and Saar (2009) argue that uninformed traders disturb the process of incorporating new information into prices. Furthermore, Beckmeyer et al. (2023) emphasize the speculative nature of 0DTE options trading, documenting both the strong preference and significant losses of retail investors on these contracts.

Consistent with our conjecture, we find that an increase in 0DTE options trading leads to higher volatility.[4] Using the staggered introduction of weekly index options as an instrument variable to overcome

[3] Trends in options trading (2023): https://www.nyse.com/data-insights/trends-in-options-trading

[4] For brevity, we use the term "*volatility*" to mean the daily standard deviation of market indices' (S&P 500, NASDAQ-100, and Russell 2000) midpoint quote return every 5 minutes between 9:30 AM and 4:00 PM Eastern Time.

endogeneity issues, we document that a one standard deviation increase in 0DTE options volume raises volatility by 9.10% relative to its mean value, and 15.91% relative to its standard deviation. Moreover, we show that higher 0DTE options trading worsens market efficiency.

Until recently, trading on 0DTE options linked to the index was not particularly popular. For example, in the S&P 500 index during January 2011, the monthly 0DTE options trading volume was approximately 80,000 contracts. This was less than 1% of the monthly options trading volume in the S&P 500 index-linked options trading volume. The introduction of SPXW has given investors more frequent short-term expiration options trading opportunities, including 0DTE options. Starting from the introduction of Friday expiration SPXW in December 2010, investors could access 0DTE options linked to the S&P 500 index at least once a week.[5] Since then, Monday, Tuesday, and Wednesday expiration options have been listed in the market, and lastly, with the introduction of Thursday expiration options in May 2022, investors have been able to trade S&P 500-linked 0DTE options every business day.[6] Similarly, weekly options on the NASDAQ-100 Index were introduced starting in 2018, followed by the Russell 2000 Index in 2024. Detailed information about each introduction of weekly index options is provided in Table 1.

[Insert Table 1 here]

Since the introduction of weekly options, 0DTE options trading has grown rapidly. The time series evolution of 0DTE options trading linked to the S&P 500, NASDAQ-100, and Russell 2000 index is presented in Figure 1.

[5] https://ibkr.info/node/1456

[6] Monday-expiration option: https://ir.cboe.com/news/news-details/2016/CBOE-to-List-SPX-Monday-Expiring-Weeklys-Options-07-11-2016/default.aspx,
Wednesday-expiration option: https://ir.cboe.com/news/news-details/2016/CBOE-to-List-SPX-Wednesday-Expiring-Weeklys-Options-02-01-2016/default.aspx,
Tuesday- and Thursday-expiration option: https://ir.cboe.com/news/news-details/2022/Cboe-to-Add-Tuesday-and-Thursday-Expirations-for-SPX-Weeklys-Options-04-13-2022/default.aspx

[Insert Figure 1 here]

Figure 1 shows a trend of average daily 0DTE options trading volume as a percentage of the total trading volume for the index-linked options, denoted by 0DTE%. On the last Friday of January 2011, the daily 0DTE trading volume was 4.31% of the total S&P 500-linked options trading volume. But, on the last Friday of August 2023, it was 52.06% of the total options trading volume. Likewise, on the same date, 0DTE options trading was 43.05% of the total options trading volume in NASDAQ-100.

Before testing the relationship between 0DTE% and volatility, we first document what drives 0DTE options trading. Following the literature (Engle and Rangel 2008; Paye 2012; Baker et al. 2016), we consider index- and macro-related variables that may affect 0DTE options trading. We find several economically important drivers. Index-related variables, such as the index return or volume, display a negative relationship with 0DTE options trading, while the index volatility of 5 business days ago exhibits a positive relationship with 0DTE options trading. In terms of macro-related variables, spread variables, such as the CP spread and the default spread, display a negative relationship with 0DTE options trading. On the other hand, the Economic Policy Uncertainty (EPU) index, the volatility of the foreign exchange rates, and that of the Consumer Price Index (CPI) show a positive relationship with 0DTE options trading.

We next consider the relationship between 0DTE% and volatility. We begin with a simple ordinary least squares (OLS) regression and regress volatility on 0DTE%. We use the daily standard deviation of the SPDR S&P 500 (SPY), Invesco QQQ (QQQ), and iShares Russell 2000 (IWM) ETF's midpoint quote return every 5 minutes between 9:30 AM and 4:00 PM Eastern Time as the main dependent variable. 0DTE% is the independent variable of interest and is defined as the proportion of 0DTE options trade volume over the total trade volume of the index-linked option on a given day, where indices are S&P 500, NASDAQ-100, and Russell 2000 index. We find a positive association between 0DTE% and volatility. A one standard deviation increase in 0DTE% is associated with an approximately 9.82% increase relative to the mean value of volatility, which is 17.17% of its standard deviation.

When using OLS estimation, endogeneity concerns about understanding the effect of 0DTE on volatility arise. Both reverse causality and omitted variable bias may contaminate the OLS regression results. In terms of reverse causality, higher volatility could attract more investors to trade 0DTE options. Moreover, an omitted variable can cause the coefficient estimates to be biased and inconsistent. News or events, such as major economic announcements that affect both 0DTE options trading and volatility, could be an omitted factor from the model. For example, a release of FOMC's beige book or the Consumer Price Index announcement could lead to increased trading activity in 0DTE options for investors to leverage the information provided in these announcements, resulting in higher volatility (Veronesi 1999; Nofsinger et al. 2003).

To overcome these endogeneity concerns, we employ an instrumental variables (IV) approach. We instrument the time *t*'s 0DTE% with the staggered introduction of weekly index options (Foucault et al. 2011; Hendershott et al. 2013). The introduction offers quasi-experimental variation in 0DTE options trading activity that is exogenous to short-term volatility shocks, which enables us to identify the causal impact of 0DTE options trading on market volatility.

We argue that our instrumental variable satisfies the necessary conditions of relevance, exogeneity, and the exclusion restriction. First, the instrument demonstrates the relevance confirmed by a significant link between the staggered introduction of weekly options and 0DTE trading volume, with a Kleibergen-Paap *F*-statistic of 3,511. Second, the exogeneity condition is met because the listing of these options reflects structural decisions by exchanges to enhance product flexibility, orthogonal to concurrent market volatility. Finally, regarding the exclusion restriction, we posit that given the exceptional depth of the underlying index markets, the addition of weekly options is unlikely to exert a significant effect on volatility via another channel, such as the liquidity channel. Therefore, the instrument influences volatility primarily through the channel of increased speculative 0DTE trading.

Using this IV design, we regress volatility on the instrumented 0DTE%. We find that 0DTE options trading has a significant positive impact on volatility, and the results are statistically and economically significant. The economic magnitude is large but reasonable: one standard deviation in the instrumented 0DTE% causes a 9.10% increase relative to the mean value of the volatility, which is 15.91% of its standard deviation.

High volatility does not necessarily mean worse market quality. A market that more quickly adapts to information or that experiences higher flows of information would also have high volatility. It could be that 0DTE% is high when there is more information, which leads to high volatility. To distinguish whether higher volatility is capturing more information or more noise, we consider two price efficiency measures: the Hasbrouck (1993) pricing error and the variance ratio (VR). We repeat the IV approach with these price efficiency dependent variables and find that increased 0DTE options trading harms market efficiency.

To additionally support our idea that retail traders drive 0DTE options trading, we exploit outages of Robinhood, a major brokerage platform for retail investors. If retail investors are the primary participants in 0DTE options market, their unexpected inability to trade during outages should lead to a measurable decline in 0DTE options trading. Consistent with this hypothesis, we find that 0DTE options trading falls by 2.5% to 5.9% during Robinhood outages, corresponding to 15.84% to 37.39% of its mean, providing evidence that retail traders are the primary investors of 0DTE options trading.

Lastly, we conduct a variety of tests to further refine the impact of 0DTE options trading on volatility. First, we restrict the sample to narrower windows around key introduction phases of weekly index options. Across all subsample specifications, we continue to find robust and consistent effects, confirming that our results are not driven by long-term trends or event timing in 0DTE options or volatility. Second, to clarify whether the volatility induced by 0DTE options trading spills over beyond the trading day, we decompose return variation into close-to-close, intraday, and overnight components. We find that 0DTE options trading raises close-to-close and intraday return variations but has only a small and statistically weak effect on overnight

returns, indicating that the impact is concentrated in same-day expirations rather than risk carried across sessions.

While we acknowledge the potential volatility-dampening channel driven by market makers' gamma hedging suggested in recent literature (Dim et al., 2023; Adam et al., 2024; Vasquez et al., 2025), our analysis demonstrates that the net impact of 0DTE options trading is an increase in market volatility. We provide a detailed reconciliation of these divergent findings, addressing methodological distinctions and controlling for hedging activities, in the Internet Appendix.[7]

This paper contributes to the extant literature on options trading and its effect on underlying assets, where mixed results are suggested. Several papers document that options trading decreases volatility in the underlying asset (Conrad 1989; Skinner 1989; Ni et al. 2021). Others find an increase in volatility of the underlying asset (Day and Lewis 1988; Harris 1989; Ni, Pan, and Poteshman 2008; Ni et al. 2021). Finally, a subset of the literature shows no relationship (Kamara, Miller, and Siegel 1992; Long, Schinski, and Officer 1994; Darrat and Rahman 1995; Muravyev, Pearson, and Broussard 2013). Our results support that options trading affects the volatility of underlying assets.

We also contribute to a growing body of research that investigates short-term expiration options, 0DTE options trading, and its impact on the market. Several papers advance theoretical understanding of 0DTE options trading (Almeida et al. 2024; Johannes et al. 2024). Another stream of literature focuses on the relationship between 0DTE options trading and volatility. Within this stream, the empirical evidence is mixed. On one hand, some studies argue for a volatility-dampening effect, positing that 0DTEs stabilize

[7] A detailed discussion regarding studies that document a volatility-dampening effect of 0DTE options trading along with additional validation tests is provided in Internet Appendix IA1. Specifically, in IA1.1 and IA1.2, we demonstrate that 0DTE trading leads to a significant net increase in volatility even after controlling for market makers' net gamma exposure and accounting for order imbalances, indicating that the destabilizing effect of speculative trading outweighs the stabilizing effect of hedging. Furthermore, IA1.3 and IA1.4 clarify that the discrepancies with these studies stem from differences in the analysis horizon (intraday vs. next-day), variable measurement, and identification strategies regarding endogeneity (e.g., the validity of DID and IV settings).

markets via market maker gamma-hedging (Dim et al. 2023; Adams et al. 2024; Vasquez et al. 2025). On the other hand, other research suggests that 0DTEs amplify volatility (Weiss et al. 2024). Our findings thus provide a more comprehensive picture, suggesting the overall impact is positive despite the existence of a potential dampening channel.

Our paper contributes to literature regarding the impact of options trading on price efficiency where mixed findings are suggested as well. Several studies document that options trading leads to better market quality and pricing efficiency (Kumar et al. 1998; Cao 1999; Cao and Ou-Yang 2008; Roll, Schwartz, and Subrahmanyam 2009; Blanco and Wehrheim 2017). More generally, several studies document that options trading volumes are informative about future stock prices by providing additional information in the market across various settings (Figlewski and Webb 1993; Easley, O'Hara, and Srinivas 1998; Chakravarty, Gulen, and Mayhew 2004; Pan and Poteshman 2006; Hu 2014; Bai, Philippon, and Savov 2016). However, a handful of papers find the opposite, that options trading harms information and efficiency (Chiras and Manaster 1978; Galai 1978; Day and Lewis 1988; Froot et al. 1992; Ni, Pearson, and Poteshman 2005). Our paper reinforces the idea that options trading is harmful to market quality.

Lastly, our paper also contributes to literature on retail investor behavior's impact on the market by examining their activity on the Robinhood platform. Several studies use Robinhood data to examine retail investors' impact on market liquidity, volatility, portfolio holdings, and attention-driven trading (Ozik et al. 2021; Welch 2022; Bogousslavsky and Muravyev 2024), while others explore broader outcomes such as ETF ownership and macroeconomic responses (Ben-David et al. 2023; Greenwood et al. 2023). Other studies exploit Robinhood outages as exogenous shocks to identify the effect of retail trading on the equity market (Barber et al. 2022; Eaton et al. 2022). We apply these events to the options market to identify the impact of retail investors on 0DTE options trading.

2. Data and Variables

In this section, we introduce the data sources and describe the construction of the variables we use. We provide detailed information on these variables' summary statistics. Then, we conduct an analysis to explore the determinants of Zero-Day-to-Expiration (0DTE) trading volume.

A. Main Variables

We mainly utilize the Trade and Quote (TAQ) and OptionMetrics databases to examine the impact of 0DTE options trading on volatility. We use the daily volatility as our main dependent variable $(\sigma_{i,t})$. To calculate the volatility measure, we utilize SPY, QQQ, and IWM's quote prices from the TAQ database from January 2010 to August 2023.[8] Specifically, we select the midpoint between the bid and ask quote prices every 5 minutes between 9:30 AM and 4:00 PM, resulting in a total of 78 midpoint observations per day. In cases where there is no quote during a given period, we use the midpoint of the previous period to ensure the continuity of the series. Using the midpoint prices, we calculate the 5-minute return and compute the standard deviation of the returns over each trading day to obtain volatility. This procedure helps the volatility measure capture the intraday price fluctuations while adjusting for the effect of a period with no quote.

The main independent variable in this study, 0DTE%, is derived from OptionMetrics data for the period spanning from January 2010 to August 2023. 0DTE% is defined as:

$$0DTE\%_{i,t} = \frac{index\ linked\ 0DTE\ options\ trading\ volume_{i,t}}{Total\ trade\ volume\ of index\ linked\ options_{i,t}}, \quad (1)$$

[8] As of March 2025, OptionMetrics provides the data available until 31st, August 2023.

where $index \in$ {S&P 500, NASDAQ-100, Russell 2000}, and $0DTE\%_{i,t}$ measures the proportion of 0DTE options volume traded on a given day relative to the total number of index-linked options traded on a given day.

When counting the number of days to expiration, we exclude weekends and holidays. For example, when an expiration date falls on a Monday, the trading on the previous Friday is counted as 1DTE to be consistent with the perspective of an investor.

B. Summary Statistics and Control Variables

We present the descriptive statistics for the main dependent variable ($\sigma_{i,t}$), independent variable ($0DTE\%_{i,t}$), and control variables in Panel A of Table 2. We provide a detailed definition and calculation of all variables in Table IA1. $\sigma_{i,t}$ is in basis points (bps). The mean value of $\sigma_{i,t}$ is 9.56 bps with a standard deviation of 5.47 bps. When annualized, the average is 13.40%, with a corresponding standard deviation of 7.67%. The average $0DTE\%_{i,t}$ is 7.37%, and the standard deviation is 13.81% during the sample period. There is a strong increasing trend for $0DTE\%_{i,t}$ over the sample, with more recent $0DTE\%_{i,t}$ values exceeding 40%. We document the time series of $0DTE\%_{i,t}$ in Figure 1. $0DTE\%_{i,t}$ was less than 5% until 2016, while it was more than 40% after 2023 on average. Panel B of Table 2 shows the correlation between the variables. All correlations are relatively low, with the largest magnitude being 0.65 between $\sigma_{i,t}$ and $\sigma_{i,t-5}$. The correlation table suggests that multicollinearity is not a significant concern. Since the stationarity of main variables could be a potential issue, we conduct autocorrelation tests such as the Augmented Dickey-Fuller test and Phillips-Perron test, and confirm that the autocorrelation is not present. For details, see Section 3. C (Validation Test).

[Insert Table 2 here]

C. What Drives 0DTE%?

As little is known about what drives 0DTE options trading activity, we first analyze its determinants. As we focus on the index-linked short-term expiration options, we consider index-level and macroeconomic variables as determinants. With respect to index-level variables, we choose the index return, index volume, and the volatility five business days ago because they are associated with the prices of index options.[9] For macroeconomic variables, we include the commercial paper (CP) spread, the default spread, the volatility of foreign exchange, the volatility of inflation, and the economic and political uncertainty (EPU) measure introduced by Baker et al. (2016). Since these variables are associated with market-level risk and uncertainty, they may impact 0DTE%. Paye (2012) argues that the CP spread and default spread can be highly associated with the business cycles and investors' uncertainty on fundamentals. Also, Engle and Rangel (2008) suggest that exchange rates and inflation are important factors in evaluating the future uncertainty in the aggregate economy. Baker et al. (2016) measure of EPU captures the movement of policy-related economic uncertainty. In the analysis, we divide EPU by 100 to make the coefficient comparable.

To examine the determinants of 0DTE options trading, we regress 0DTE% on the index-level and macroeconomic control variables as the independent variables. The interpretation of the estimated coefficients is that a standard deviation change in an independent variable is associated with a beta-standard deviation change in the dependent variable.

The estimated equation is:

$$0DTE\%_{i,t} = \alpha + \beta' X_{i,t} + \omega_i + \gamma_t + \varepsilon_{i,t}, \quad (2)$$

[9] We divide the index volume by 1,000,000 to make the coefficient comparable.

where $0DTE\%_{i,t}$ is computed as described in equation (1), $X_{i,t}$ is a vector of control variables, ω_i is an index fixed effect, and γ_t is a day of the week fixed effect. The standard errors are clustered at the daily level.

Table 3 presents the determinants of 0DTE options trading. We provide four distinct outcomes, each reflecting a combination of including or excluding index or days of week fixed effect and clustered standard error at the daily level. Regardless of these combinations, the results are consistent across all specifications. We adopt the inclusion of fixed effects and the clustered standard errors at the daily level as the standard approach for the remainder of the paper to provide the most conservative results.[10]

We find that the determinants account for around 28.6% of the variation in 0DTE options trading when including the fixed effect, which is presented in columns (2) to (4). Regarding index-level variables, we find that both index return and volume exhibit a negative relationship with 0DTE%, while the volatility of the index on the same day in the previous week exhibits a positive relationship with 0DTE%. With respect to macroeconomic variables, overall, we find that all spread variables have a negative relationship with 0DTE%. It implies wider spreads are associated with lower 0DTE options trading. Additionally, 0DTE options trading is more responsive to the default spread than the CP spread. EPU has a positive relationship with 0DTE%, implying that overall market uncertainty is highly related to 0DTE options trading activity. The volatility of the foreign exchange rates and inflation also has a positive relationship with 0DTE%.

[Insert Table 3 here]

[10] The results are robust to different time fixed effects and to clustered standard errors at different time levels, as detailed in Table IA8 and IA9.

3. Does 0DTE Options Trading Increase Volatility?

In this section, we test whether 0DTE options trading increases volatility. We first present the empirical results from an ordinary least squares (OLS) regression. To overcome endogeneity issues, we implement an instrumental variables (IV) approach. Finally, we test whether price efficiency is harmed when trading more 0DTE options. We find that 0DTE options trading leads to higher volatility and worse price efficiency.

A. Relation between 0DTE Options Trading and Volatility

We start examining the relationship between 0DTE options trading and volatility by documenting its contemporaneous movements. We regress $\sigma_{i,t}$ on $0DTE\%_{i,t}$ and control variables that correspond to the same given day.

The estimated equation is:

$$\sigma_{i,t} = \alpha + \beta\ 0DTE\%_{i,t} + \delta' X_{i,t} + \omega_i + \gamma_t + \varepsilon_{i,t}, \quad (3)$$

where $\sigma_{i,t}$ is the daily standard deviation of an index's midpoint quote return every 5 minutes, $0DTE\%_{i,t}$ is computed as described in equation (1), $X_{i,t}$ is a vector of control variables, ω_i is an index fixed effect, and γ_t is a day of the week fixed effect. The standard errors are clustered at the daily level. Table IA1 provides a comprehensive summary of the definition and construction methods of variables.

Table 4 reports the results of the regression. Column (1) in Table 4 is a simple univariate regression and reveals a positive and statistically significant relationship between 0DTE% and volatility, with a coefficient of 0.068, significant at the 1% level. Thus, a one standard deviation increase in 0DTE% is associated with a 9.82% increase relative to the mean value of volatility, which is 17.17% of its standard deviation. Furthermore, the inclusion of control variables in the regression models strengthens the robustness of these findings. Column (2) is a regression with index-level control variables, such as the index return and index

volume. To alleviate the concerns that the volatility of the index might have a strong autocorrelation and time trends, we further include the volatility of the index five business days ago. The estimated coefficient still displays a positive and significant relation between 0DTE% and volatility. Column (3) includes a full set of control variables. It incorporates additional macro-related factors such as spread measures, the volatility of foreign exchange and CPI, and EPU. The result indicates that the positive and significant relationship between 0DTE% and volatility persists.[11] The estimated coefficient is 0.056 and significant at a 1% level. It indicates that a one percentage point increase in 0DTE% is associated with a 0.056 bp increase in volatility. Given that the standard deviation of 0DTE% is 13.81 (in percentage), and the coefficient of 0DTE% is 0.056, the result shows that a one standard deviation increase in 0DTE% is associated with an approximately 8.09% increase relative to the mean value of the volatility, which is 14.14% of its standard deviation.

[Insert Table 4 here]

B. Effect of 0DTE Options Trading on Volatility

There is a potential endogeneity issue that could contaminate the interpretation of the OLS results. The endogeneity problem arises from the possibility of reverse causality between the 0DTE options trading and volatility. Changes in volatility could affect 0DTE%. For example, higher volatility might lead investors to trade more with 0DTE options. The reverse causality concern may also arise in the presence of autocorrelation of volatility (Ding, Granger, and Engle 1993; Bollerslev and Mikkelsen 1996; Paye 2012). Time *t*'s volatility is influenced by past volatility values, which can create an endogeneity issue. The OLS results can also be contaminated by omitted variables that may affect both the trading activity of 0DTE

[11] Additionally, we perform an analysis whereby all control variables are lagged by one period, replacing $X_{i,t}$ with $X_{i,t-1}$ in equation (3). The results are economically consistent with the specification using contemporaneous control variables. We provide these results in Table IA10. We perform the same robustness checks for Table 5.

options and volatility. For example, the release of the FOMC's beige book or CPI announcement may lead to a surge in trading activity in 0DTE options as investors leverage the information from these announcements. These announcements also affect volatility (Veronesi 1999; Nofsinger et al. 2003).

To establish a causal link between 0DTE options trading and volatility, we employ an IV approach that addresses endogeneity concerns arising from reverse causality and omitted variable bias (Wooldridge 2010). We use the staggered introduction of weekly index options for our instrument variable. Weekly index options are introduced for the S&P 500 and NASDAQ-100, but not for the Russell 2000, during our sample period. This introduction timing difference is an ideal setting for our tests. Once weekly options are introduced, we consider those indices as the treated group, and we consider other indices before the introduction as the control group. More specifically, for the S&P 500 and NASDAQ-100, we set index–weekday treatment indicators equal to one on the date when weekly options are first listed and remain one thereafter, and zero otherwise. Accordingly, each pair of index–day of the week becomes treated at a different calendar time. By contrast, the Russell 2000 first introduced their weekly options in January 2024, which is after our sample period, so the Russell 2000 effectively serves as a control group in our specifications.

The variation in introduction timing further supports causal inference, as it was driven by exchanges' operational decisions independent of index-specific volatility or demand shocks. Moreover, in line with Hendershott et al. (2013), we include both time and index fixed effects that allow us to absorb market-level changes in volatility and isolate the differential impact of the introduction across treated and not-yet-treated indices.

This instrument is more suitable than an alternative candidate, which is the earlier introductions of standard index options, because it aligns more closely with the objective of our study. Traditional index options on the S&P 500 and NASDAQ-100 offered limited opportunities for 0DTE trading, with expirations typically occurring quarterly or, in some cases, only once every few years. In contrast, the introduction of weekly

index options substantially increased the accessibility of 0DTE trading by allowing for daily expirations. This structural change makes a fundamental shift in how investors engage with short-term options and coincides with the recent surge in 0DTE activity. Therefore, using the introduction of SPXW and NDXP helps us to identify the effects of frequent access to 0DTE options on the market.

For our instrumental variable to be valid, it must satisfy the exclusion restriction, exogeneity, and relevance conditions. We argue that these conditions are likely to hold in our setting. With respect to the exclusion restriction, it is required that the introduction of weekly index options cannot correlate with the volatility, but through its relation with the time *t*'s 0DTE%. The primary challenge to this assumption is the potential for an alternative channel. For instance, increased availability of short-term options could encourage hedging for market makers and institutional investors, thereby improving the liquidity of the underlying index, which in turn could dampen volatility.

However, we argue that this liquidity channel is unlikely to be a primary driver of the volatility effects we observe. The underlying index market (e.g., S&P 500 and NASDAQ-100) is already exceptionally deep and liquid. It is therefore improbable that the marginal addition of weekly options would substantially alter the fundamental liquidity characteristics of such a vast market. Rather, the most direct and pronounced effect of introducing these instruments is to expand the accessibility and trading volume of short-term options that generates a speculative trading-friendly environment. Therefore, it is likely that any observed changes in market volatility following the staggered introduction of weekly options should stem from its effect on 0DTE trading activity, ensuring the validity of the exclusion restriction in our analysis.

To satisfy the exogeneity condition, the introduction of weekly index options is not directly influenced by short-term market volatility. Because the timing of weekly option introductions was determined by exchanges with the goal of enhancing product flexibility and market liquidity rather than in response to

movements in index-level volatility[12], the IV allows us to hold the exogeneity condition. The staggered introduction across different indices and days of the week further supports the claim that the introduction is orthogonal to contemporaneous volatility conditions.

In terms of the relevance condition, we examine the Kleibergen-Paap *F*-statistic from the first stage of the IV regression that explains the power of IV as well as the coefficient on the $introduction_{i,t}$ variable. The estimated equation for the first-stage regression is:

$$0DTE\%_{i,t} = \alpha + \beta\ introduction_{i,t} + \delta' X_{i,t} + \omega_i + \gamma_t + u_{i,t}, \quad (4)$$

where $0DTE\%_{i,t}$ is computed as described in equation (1), $introduction_{i,t}$ is the dummy variable set to zero before the introduction of weekly index options and one afterward, $X_{i,t}$ is a vector of control variables, ω_i is an index fixed effect, and γ_t is a day of the week fixed effect. The standard errors are clustered at the daily level.

We find that the IV is statistically significant, supported by the Kleibergen-Paap *F*-statistics of 3,511 when we incorporate all control variables in equation (4).[13] Furthermore, as we will discuss in detail below, the significant coefficient from the first stage regression shows that our instrument is related to the main independent variable, 0DTE%. Thus, the IV approach using the staggered introduction of weekly index options as an instrument provides us with a consistent estimate to measure the causal impact of 0DTE% on volatility.

To further validate the first-stage regression of our IV strategy and to assess the parallel-trends assumption in the staggered introduction design, we provide visual evidence in Figure 2, which plots the 0DTE options trading around the introduction of weekly index options. In Figure 2, we align each index to its own

[12] https://www.cboe.com/insights/posts/the-evolution-of-same-day-options-trading/

[13] Kleibergen-Paap *F*-statistic is proposed by Kleibergen and Paap (2006) to evaluate the overall strength of the instruments in the first-stage regression. Our results are robust with Cragg and Donald (1993) *F*-statistic. Given that the Stock and Yogo (2005) 10% threshold is 16.38, the *F*-statistic implies that the instrument is sufficiently strong.

introduction date and stack observations at the weekly level before and after a 10-week window, then average the series separately for treated and never-treated indices. The figure shows that 0DTE option trading remains flat and near zero during the pre-introduction period but rises sharply immediately after the introduction, followed by a gradual increase over subsequent weeks. This pattern confirms that the surge in 0DTE options trading is tightly concentrated around the option introduction event. Accordingly, this visual evidence supports a clean first stage, shows no differential pre-trends, and reinforces the validity of using the staggered introduction of weekly options as an instrument for 0DTE options trading.

[Insert Figure 2 here]

Using the valid IV, we estimate the instrumented 0DTE% by regressing the independent variable, 0DTE%, on the staggered introduction of weekly index options in the first stage of IV regression. After getting the instrumented 0DTE%, we regress volatility on the instrumented 0DTE% in the second stage of IV regression.

The estimated equation for the second stage regression is:

$$\sigma_{i,t} = \alpha + \beta\ 0\widehat{DTE\%}_{\iota,t} + \delta' X_{i,t} + \omega_i + \gamma_t + \varepsilon_{i,t}, \tag{5}$$

where $\sigma_{i,t}$ is the daily standard deviation of an index's midpoint quote return every 5 minutes, $0\widehat{DTE\%}_{\iota,t}$ is the estimated 0DTE% from the first-stage regression in equation (4), $X_{i,t}$ is a vector of control variables, ω_i is an index fixed effect, and γ_t is a day of the week fixed effect. The standard errors are clustered at the daily level.

Table 5 presents the results of the IV regressions. Columns (1) to (3) report the results of various specifications of the first stage IV regressions. The findings reveal that the IV satisfies the relevance of the instrument assumption, as evidenced by the statistically significant coefficients regardless of the inclusion

of control variables. For example, column (1) shows the results from a simple univariate regression in equation (4). The estimated coefficient of IV is 27.268 and is significant at a 1% level.

Next, we include several control variables in regressions, and the finding is consistent that 0DTE% at time *t* is highly correlated with the staggered introduction of weekly index options. In column (2), we include index-level variables. Even after controlling for index-related factors, the result is still significant. The coefficient is 26.406 and significant at a 1% level. Column (3) shows the results from regression with full specification in equation (4), which further includes macro-related control variables. The coefficient is 21.946 and is still significant at a 1% level. It implies that the 0DTE% exhibits a high degree of inertia and remains relatively unchanged over the time horizon. The validity of the IV is also supported by the Kleibergen-Paap *F*-statistic of 3,511 when incorporating all relevant control variables. Given that the Stock and Yogo (2005) 10% threshold is 16.38, the *F*-statistic implies that the instrument is sufficiently strong.

Columns (4) to (6) in Table 5 report the results of the second stage of IV regressions. For an instrumented 0DTE% ($0\widehat{DTE\%}_{i,t}$) in the second-stage regression, we employ the estimate in column (3) obtained from the first-stage regression with all relevant control variables.[14] The results show that, regardless of control variable sets, there is a statistically significant and positive relationship between the estimated 0DTE% and volatility at a 1% level.[15] Column (4) shows the result of univariate regression in equation (5). The estimated coefficient is 0.069, and it is statistically significant at a 1% level. Column (5), which includes index-level control variables, also shows that the instrumented 0DTE% positively affects volatility at a 1% level. Moreover, in column (6), which incorporates all relevant control variables, including macro-related

[14] The results obtained from different instrumented 0DTE% measures, using alternative specifications with different control variable sets in the first-stage regression, consistently align with our findings.

[15] To mitigate heteroskedasticity in the volatility and 0DTE options trading at different times, we provide results with alternative clustered standard errors in Table IA9. The results are robust across alternative clustering choices. We perform the same robustness checks for Table 5.

variables, the coefficient is 0.063 and statistically significant at the 1% level. [16] The results show that a one standard deviation increase in 0DTE% causes a 9.10% increase relative to the mean value of the volatility, which is 15.91% increase relative to its standard deviation. The results suggest that an increase in 0DTE options trading leads to increased volatility.

Our results align with the findings in several prior studies that options trading results in an increase in volatility. Day and Lewis (1988) document that the option prices are positively associated with the volatility of the underlying stocks, specifically around both quarterly and non-quarterly expiration dates. Harris (1989) finds that the volatility of the S&P 500 index is heightened after the introduction of futures and options products on the S&P 500 index. Also, Ni, Pan, and Poteshman (2008) show that the non-market makers' demand for volatility in the options market is positively related to the subsequent realized volatility of underlying stocks. In this regard, our paper reinforces the results of previous literature that options trading leads to an increased volatility of underlying assets. Additionally, our paper supplements this literature by particularly focusing on short-horizon options trading and showing that 0DTE options trading increases volatility.

[Insert Table 5 here]

C. Validation Tests

[16] The first stage shows that the staggered introduction of weekly index options increases 0DTE options trading by 21.946, which is significant at the 1% level. We do not include in the table, but the reduced form regresses volatility directly on the same introduction indicators and shows that these events increase volatility by 1.380, also significant at the 1% level. And we document that the coefficient of our second-stage estimates in column (6) is 0.063 and significant at the 1% level. The implied ratio of the reduced form to the first stage, 1.380/21.946≈0.063, matches the second stage coefficient reported in column (6) of Table 5, which is 0.063 and significant at the 1% level.

There are several econometric concerns regarding equation (5) that would require further evaluation. We, therefore, evaluate whether 0DTE% is stationary; whether the dependent variable exhibits autocorrelation; and whether the timing of weekly option introductions may not be fully random.

Non-stationary independent variables can lead to spurious regression results, suggesting a relationship between variables even when there is none. Figure 1 shows an increasing trend in daily 0DTE% from January 2010 onwards, raising doubts about whether it is stationarity. To resolve the doubts, we conduct tests for stationarity in 0DTE% using the Augmented Dickey-Fuller test and Phillips-Perron test (Plazzi, Torous, and Valkanov 2010; Golez and Koudijs 2018; Chen, Joslin, and Ni 2019; Huang and Kilic 2019; Ranaldo, Schaffner, and Vasios 2021). Both test results reject the presence of a unit root at the 1% significance level. The test statistics of the Augmented Dickey-Fuller test and Phillips-Perron test for our variable are -52.418 and -61.161, respectively, below the critical value at the 1% level for both tests.[17] The results confirm the stationarity of the 0DTE% variable.

Additionally, we plot the daily time-series trend of the fitted value of 0DTE% ($0\widehat{DTE\%}_{\iota,t}$) in Figure 3. This variable is the estimate from the first-stage of IV regression from equation (4), incorporating all control variables, the index fixed effect, and the day of the week fixed effect. This figure provides illustrative evidence that our main variable, 0DTE%, exhibits stationarity when instrumented with the staggered introduction of weekly index options. In addition, both the Augmented Dickey-Fuller and Phillips-Perron tests reject the presence of a unit root at the 1% significance level. Specifically, the test statistics for the instrumented 0DTE% are -10.242 and -7.712, respectively, both well below the critical value at the 1% level, indicating strong evidence of stationarity. Therefore, this evidence alleviates the stationarity concerns of 0DTE%.

[Insert Figure 3 here]

[17] The critical values at 10%, 5%, and 1% levels are -2.57, -2.86, and -3.43, respectively, for both tests.

A second potential concern is autocorrelation in volatility. Past volatility may predict time *t*'s volatility as stock return volatility is persistent (Poterba and Summers 1986; French, Schwert, and Stambaugh 1987; Schwert 1989). Although we first control the volatility of the index five business days ago, which is $\sigma_{i,t-5}$, given the strong volume–volatility relation, it may not be sufficient to control for lagged index volatility only one or five days ago. To address this concern, we re-estimate both the OLS (i.e., Table 4) and IV (i.e., Table 5) specifications by including up to five lags of realized volatility in Table IA12. This control variable helps absorb predictable patterns in volatility that may stem from weekly cycles, allowing us to better isolate the variation attributable to 0DTE options trading activity. Our results remain robust to additional volatility control variables.

The next concern is that the timing of new option introductions may not be fully random and could be correlated with underlying market conditions. To examine this possibility, we implement an event-time analysis that allows us to detect potential pre-trends and assess how volatility responds around the introduction events. Figure 4 reports the mean difference in volatility between treated and control indices by stacking weekly observations within a four-week window before and after each introduction. The event-time coefficients during the pre-treatment period are close to zero and statistically insignificant, indicating no systematic differences in volatility leading up to the introduction events. In contrast, the post-treatment coefficients rise gradually and become statistically significant only several weeks after the introduction, with the bars moving away from zero after roughly the third week. This delayed response suggests that although the option products became available immediately, market participants likely needed time to recognize and actively trade the new instruments. Overall, the absence of pre-trends and the gradual emergence of post-treatment effects support the view that volatility responds to increased 0DTE options trading rather than to anticipation effects or endogenous introduction timing.

[Insert Figure 4 here]

D. Effect of 0DTE Options Trading on Price Efficiency

This section examines whether 0DTE options trading is harmful to the price efficiency of the market. Increased volatility led by increased 0DTE options trading would mean either better or worse price efficiency. If increased volatility is driven by quickly adapting information on 0DTE options trading, the high volatility may indicate improved price efficiency. However, if 0DTE options trading leads to increased volatility due to retail investors who are characterized by their noise trading or speculative investment behaviors, it is likely to undermine price efficiency, as suggested by theoretical predictions. For example, such traders may introduce more noise into the market, leading to less informative prices and lower price efficiency in the market (Chiras and Manaster 1978). Hence, based on the theoretical prediction and the prevalence of retail investors in the 0DTE market, we hypothesize that the increase in 0DTE options trading is detrimental to the price efficiency of the market. We test this hypothesis through Hasbrouck's pricing error tests.

We employ Hasbrouck's (1993) pricing error to examine the association between 0DTE options trading and the price efficiency of the market. Hasbrouck (1993) proposes a model where the observed transaction price can be broken down into two components: an efficient price and a pricing error. The efficient price represents the expected value of the security based on all available information at the time of the transaction. It is assumed that the stock price follows a random walk and only changes when new information becomes available. In contrast, the pricing error accounts for deviations between the observed and efficient prices, which capture various market frictions unrelated to information. The pricing error is characterized as having a constant mean, and its properties do not change over time (i.e., zero-mean covariance-stationary process). Since the expected value of the pricing errors is zero, the standard deviation of the pricing error denoted as $V(s)_{i,t}$, serves as a measure to assess the extent of deviations from the efficient price in his model. We posit that 0DTE% is positively associated with the standard deviation of the pricing error, $V(s)_{i,t}$, because 0DTE options trading reduces the price efficiency of the market due to retail investors' speculative trading.

In the empirical estimation, we employ the methodology suggested by Hasbrouck (1993) and Boehmer and Wu (2013) to calculate the standard deviation of the price error within a daily window. We compute a daily price error, which is denoted as $\sigma_{i,t}^{HB}$, using second-by-second data from the TAQ, where the trade direction is determined by Lee and Ready's (1991) algorithm.

The estimated equation is:

$$Y = \alpha + \beta 0\widehat{DTE\%}_{i,t} + \delta' X_{i,t} + \omega_i + \gamma_t + \varepsilon_{i,t}, \quad (6)$$

where Y takes one of our main dependent variables: Hasbrouck's pricing error at t or t+1, $\sigma_{i,t}^{HB}$ or $\sigma_{i,t+1}^{HB}$, respectively. $0\widehat{DTE\%}_{i,t}$ is the estimated 0DTE% from the first-stage regression in equation (4). $X_{i,t}$ is a vector of control variables, ω_i is an index fixed effect, and γ_t is a day of the week fixed effect. The standard errors are clustered at the daily level.

Table 6 reports that 0DTE options trading positively affects price inefficiency measures in all specifications, implying that 0DTE options trading reduces the market price efficiency.[18] Columns (1) and (2) report the results from the contemporaneous model, which uses $\sigma_{i,t}^{HB}$ as a price inefficiency measure. Specifically, column (1), which includes index-level control variables, shows that the estimated coefficient is 0.246, and it is interpreted that a one standard deviation increase in 0DTE% causes a 6.10% increase in pricing errors ($\sigma_{i,t}^{HB}$) relative to their mean value, which 6.75% of its standard deviation. Column (2), with the macro-related control variables altogether, shows that the estimated coefficient is 0.241, indicating that a one standard deviation increase in 0DTE% causes a 5.97% increase in pricing errors relative to their mean value, which is 6.59% of its standard deviation. To mitigate endogeneity concerns, we also employ a lagged model ($\sigma_{i,t+1}^{HB}$) in columns (3) and (4), following Boehmer and Wu (2013). The results remain robust. Column (3) reports a coefficient of 0.222, representing a 5.50% increase in price inefficiency relative to the mean, which

[18] Our results are robust when we use OLS regression instead of IV regression in Table 6.

is 6.09% of its standard deviation. Similarly, the result in column (4) yields a coefficient of 0.219, corresponding to a 5.42% increase in price inefficiency relative to its mean, which is 5.99% of its standard deviation. Therefore, the empirical test using Hasbrouck's (1993) pricing error reinforces our hypothesis that an increase in 0DTE options trading deteriorates the price efficiency of the market.[19]

Our result is consistent with the evidence found in several previous literature that options trading destabilizes price informativeness and is detrimental to price efficiency. Galai (1978) documents that the trading strategy using options closer to their maturity is strongly related to price inefficiency. Day and Lewis (1988) show that the information transmitted from options trading around their expiration dates is positively associated with the unanticipated parts of the underlying asset's volatility, indicating the price inefficiency of underlying assets. Ni, Pearson, and Poteshman (2005) provide evidence that the clustering behavior in the delta-hedge rebalancing or stock price manipulation by investors destabilizes the prices of underlying stocks. Our results contribute to this body of research in that we confirm the negative impact of options trading on price efficiency, and in that we particularly study a new dimension of options, the short-horizon options trading.

[Insert Table 6 here]

4. Do Retail Investors Drive 0DTE Options Trading?

In this section, we further test whether retail investors are key drivers of 0DTE options trading. If retail investors are the primary participants in 0DTE options trading, we would expect 0DTE options trading volume to decline when they are not able to trade due to unavoidable circumstances. To test this hypothesis,

[19] For robustness, we also employ the Variance Ratio (VR) test to examine the impact of 0DTE options trading on price efficiency, as detailed in Section IA2.1 of the Internet Appendix. The robustness test also shows the consistent results that the impact of 0DTE options trading deteriorates price efficiency.

we exploit outages of Robinhood, a major brokerage platform for retail investors, to examine how unexpected trading interruptions to retail investors impact 0DTE options trading. Our results show that 0DTE options trading declines by 2.5% to 5.9% during Robinhood outages, suggesting that retail investors' participation plays a substantial role in driving 0DTE options trading.

A. Robinhood Outages

Robinhood has become one of the most popular platforms for retail investors. During the first quarter of 2020, Robinhood reported that its users executed trades at a rate nine times higher than E-TRADE customers and 40 times higher than Schwab clients on a per-dollar basis, highlighting the platform's unique concentration of active retail participants.[20] Furthermore, according to information released by Robinhood, as of late 2024, the platform had 24.8 million funded accounts, further demonstrating its widespread use among retail investors.[21] Regarding retail investors' 0DTE options trading, retail participation through Robinhood surged, with these contracts making up a record 56% of S&P 500 index options.[22]

Because of this retail-focused characteristic, Robinhood's data has frequently been used in the Finance literature as a proxy for retail investors' trading behavior. For example, Ozik et al. (2021) use Robinhood data to analyze stock liquidity in equity markets, while Welch (2022) examines retail portfolio construction and holding patterns using Robinhood data. Barber et al. (2022) and Eaton et al. (2022) investigate how Robinhood outages affect attention-induced trading and market quality, respectively. More recently, Ben-David et al. (2023), Greenwood et al. (2023), and Bogousslavsky et al. (2024) extend their application to areas like ETF ownership, fiscal stimulus-response, and behavioral finance by using Robinhood data. These

[20] https://www.nytimes.com/2020/07/08/technology/robinhood-risky-trading.html
[21] https://investors.robinhood.com/news-releases/news-release-details/robinhood-markets-inc-reports-november-2024-operating-data
[22] https://www.bloomberg.com/news/articles/2025-03-03/zero-day-options-hits-trading-record-on-trump-turmoil-robinhood-push

features position Robinhood as a representative and empirically robust proxy for retail investor behavior in financial markets.

Since the launch of the Robinhood platform, several unexpected outages have occurred due to unknown causes. During these outage periods, retail investors' trading activities were involuntarily halted. We leverage these trading outages to gauge the impact of retail investors on 0DTE options trading, following Barber et al. (2022) and Eaton et al. (2022). These outage events serve as a natural experiment, as most of the outages occurred unexpectedly and were not anticipated by market participants.

We identify Robinhood outages from their website, which provides the complete list of outage history.[23] From this website, we collect detailed information on each Robinhood outage, including the start and end times as well as the type of outage. Between 2015 and 2024, 108 outages have been reported. To use this outage data, we restrict our analysis to outages that occur during regular trading hours, as outages outside of trading hours would not affect investors' trading behavior. Next, we exclude all scheduled outages that are not exogenous or expected events to retail investors. After this filtering process, we identify overall 40 Robinhood outages after 2018 when Robinhood started to provide options trading. Since 2018, the average 0DTE% across the S&P 500, NASDAQ-100, and Russell 2000 index on days with Robinhood outages is 9.13%, compared to 18.47% on non-outage days over the same period (not tabulated). This difference provides preliminary evidence that retail investors account for a substantial share of 0DTE options trading. Figure 5 presents the average 0DTE% surrounding Robinhood outages for the S&P 500, NASDAQ-100, and Russell 2000 index. The figure shows an average of 0DTE% for the two trading days before and after 40 outage events. It also provides illustrative evidence that 0DTE% declines during Robinhood outages, suggesting that retail traders play a significant role in driving 0DTE options trading activity.

[23] https://status.robinhood.com/history

[Insert Figure 5 here]

B. The effect of Robinhood Outages on 0DTE Options Trading

On March 2, 2020, thousands of Robinhood traders found themselves locked out of their accounts, unable to execute trades during one of the most volatile market days in history. Such outages provide a natural setting to assess how involuntary interruptions to retail trading activity affect 0DTE options activity. To investigate whether retail investors drive the 0DTE options trading, we use the Robinhood outage that occurs during trading hours after 2018, when Robinhood started to provide options trading on its platform, as the shock to the retail investors' trading activity on 0DTE options.

During our sample period, however, Robinhood allowed options trading for individual stocks and ETFs such as SPY and QQQ but did not provide access to index options like SPXW. Therefore, we consider the cross-market transmission channel. We posit that when Robinhood experiences outages, its users lose access to 0DTE options on ETFs, which can reduce their 0DTE options trading activity. Then, delta-hedging demand of market makers who typically hedge these retail exposures using index options like SPXW or NDXP is reduced, causing 0DTE options trading to decline. This idea aligns with Dim et al. (2023), who document that SPY 0DTE options are particularly popular among retail traders, and projected delta-hedging flows are substantially larger for SPXW 0DTEs than for longer-dated options. To empirically test this cross-market transmission channel, we regress 0DTE options trading on a Robinhood outage indicator. The estimated equation is:

$$0DTE\%_{i,t} = \alpha + \beta_1\, Outage_t + \delta' X_{i,t} + \omega_i + \gamma_t + \varepsilon_{i,t}, \tag{7}$$

where $0DTE\%_{i,t}$ is computed as described in equation (1), $Outage_t$ is a dummy variable set to one if Robinhood has an outage during trading hours and zero otherwise, $X_{i,t}$ is a vector of control variables, ω_i

is an index fixed effect, and γ_t is a day of the week fixed effect. All standard errors are clustered at the daily level.

Our results support our hypothesis that retail-driven 0DTE options trading plays an important role in 0DTE option markets. Table 7 reports the results of Robinhood outages on 0DTE options trading. Column (1), which includes all identified outages, shows that 0DTE trading on index options falls by 2.552% on outage days relative to non-outage days and is statistically significant at a 5% level. Next, to address concerns that extremely short outages may not effectively disrupt retail trading, columns (2) through (4) restrict the sample to outages lasting at least 30, 60, and 120 minutes, respectively. The estimated declines in 0DTE options trading become larger in magnitude from 4.534 to 5.903% in columns (2) and (3) and remain statistically significant at 1% level, which accounts for 25% and 32.55% of the standard deviation of 0DTE%, respectively. When we restrict the outage duration to more than 120 minutes in column (4), the estimated decline in 0DTE options trading is 4.311% and statistically significant at 5% level, accounting for 23.77% of the standard deviation of 0DTE%.

These patterns reinforce our argument that 0DTE activity is largely driven by retail behavior, since sizable outages reduce retail investors' access to 0DTE options trading on ETFs such as SPY or QQQ, thereby lowering market makers' hedging demand in 0DTE trading on index options. In contrast, column (5) focuses on outages lasting less than 30 minutes. Consistent with our conjecture, the estimated effect is small and statistically insignificant, suggesting that brief disruptions are unlikely to impede retail traders' 0DTE options activity or alter dealers' hedging needs. Therefore, it supports the growing empirical evidence that retail investors account for a substantial share of short-dated option volume (CBOE 2025, Bryzgalova et al. 2023, and NYSE 2023), and with Dim et al. (2023), who show that SPY 0DTE trading is retail-dominated and generates large projected hedging flows in SPXW.

[Insert Table 7 here]

5. Robustness

In this section, we present several robustness tests. We assess whether our findings remain consistent across different subsample periods to address concerns about confounding effects from long-run trends or time-series issues in volatility. In addition, we clarify whether the volatility induced by 0DTE options trading spillovers beyond the trading day. Our results show that our results remain robust across multiple subsample windows around the key introduction phases, suggesting that the observed effects are not driven by specific time periods. We also find that 0DTE options trading raises close-to-close and intraday volatility but has only a small and statistically weak effect on overnight volatility, indicating that the impact is concentrated in same-day expirations.

A. Subsample Analysis

Our main analysis uses a long sample period spanning from January 2010 to August 2023. One potential concern from this sample period is that this extended time window may introduce confounding effects due to upward time-series trends in 0DTE options trading or autocorrelation in volatility. To address this concern, we conduct a series of subsample analyses that restrict the time window around the introductions. Table 8 reports the estimates using five alternative event windows centered around each index's introduction of weekly options. Specifically, we consider subsample periods of 1, 2, 3, 6, and 12 months before and after each introduction phase, presented in columns (1) through (5), respectively. These different event windows allow us to assess whether our baseline results are driven by some short-term dynamics around introductions or by broader shifts in market conditions in the long-term period.

Across all specifications, the coefficient on 0DTE% remains positive and statistically significant across all horizons. The magnitude of the effect tends to grow as the window widens from one to six months in columns (1) to (4), and it remains consistently strong even in the full twelve-month window in column (5). This pattern suggests that the volatility response becomes more pronounced as markets gradually adapt to the introduction of 0DTE options. Then, the effect appears to stabilize over the 12-month window, consistent with the view that the market would reach a new equilibrium following the initial adjustment period. Therefore, this evidence suggests that the relationship between 0DTE options trading and volatility is not an artifact of specific event-window choices. Instead, the effect is stable across windows of different lengths, suggesting that our results are unlikely to be driven by transient shocks, slow-moving macroeconomic conditions, or sample-selection concerns.

[Insert Table 8 here]

B. Spillover Effects

Lastly, we examine whether the volatility associated with 0DTE options trading spills over beyond the intraday window to better clarify the impact of 0DTE options trading. Following Lou et al. (2019), who emphasize that overnight and intraday returns are driven by different forces, we construct close-to-close, open-to-close (intraday), and close-to-open (overnight) returns. Then, we use absolute and squared returns for three horizons to proxy the volatility measure. This decomposition allows us to assess if the impact of 0DTE options trading is confined to the same-day expiration or propagates into the next day. If open-to-close volatility rises with 0DTE options trading while overnight volatility remains largely unaffected, this would support our view that the effect is tied to same-day expirations. By contrast, if some volatilities were to spill over into the overnight window, it would suggest other driving forces, for instance, dealer rebalancing at the close carries risk into the next session. To examine this hypothesis, we re-estimate our

baseline IV specification in equation (5), replacing the volatility measure with each of the absolute and squared return measures for those three horizons as the dependent variable.

Table 9 reports the estimate using two volatility measures for three different horizons. Columns (1) to (3) report the results using absolute returns. For close-to-close measure ($|Ret_{i,t}^{CC}|$) in column (1), the coefficient on 0DTE% is 0.0056 and statistically significant at a 1% level, and the coefficient of the intraday (open-to-close) measure ($|Ret_{i,t}^{OC}|$) in column (2) is 0.0049, statistically significant at a 1% level. By contrast, the coefficient of overnight measure ($|Ret_{i,t}^{Overnight}|$) in column (3) is only 0.0018 and statistically significant at 5% level, which is roughly one third of the close-to-close and intraday return variations, suggesting that the dampened overnight effects. Similarly, columns (4) to (6) repeat the analysis using squared returns. For close-to-close measure ($\left(Ret_{i,t}^{CC}\right)^2$) in column (1), the coefficient on 0DTE% is 0.0002 and statistically significant at a 1% level, and the coefficient of the intraday (open-to-close) measure ($\left(Ret_{i,t}^{OC}\right)^2$) in column (2) is 0.0001, statistically significant at a 1% level. By contrast, the coefficient of overnight measure ($\left(Ret_{i,t}^{Overnight}\right)^2$) in column (3) is statistically insignificant.

These results imply that the volatility associated with 0DTE options trading is concentrated within the same-day expirations because speculative investors tend to avoid overnight risk. The effect is strongest for open-to-close and close-to-close returns and does not meaningfully carry over into the overnight window. This evidence is consistent with our interpretation that the elevated volatility is more likely driven by intraday trading activity, rather than through dealer rebalancing at the close that would transmit risk into the next session.

[Insert Table 9 here]

6. Conclusion

In recent years, there has been a significant increase in the trading volume and proportion of Zero-Day-to-Expiration (0DTE) options linked to the major index, such as the S&P 500, NASDAQ-100, and Russell 2000, after the introduction of weekly index options. Since these options expire on the same day they are traded, they ensure that investors are not exposed to overnight risk and allow for the rapid realization of profit or loss. These characteristics are preferred by retail investors who favor speculative investments that would infuse noise into the price. According to a CBOE report in 2025 and a NYSE report in 2023, more than half of the trading volume in 0DTE options trading is attributed to retail investors. The growth of 0DTE option trading, led by retail investors, has raised concerns about the impact of 0DTE options on volatility, potentially leading to a reduction in market efficiency. The absence of prior research regarding the effect of 0DTE options on the financial market has prompted us to investigate the relationship between 0DTE options trading and volatility.

Adopting an instrumental variable approach to address endogeneity concerns, we examine the causal impact of 0DTE options trading on volatility. We find that an increase in 0DTE options trading results in higher volatility. Furthermore, we find that 0DTE options trading deteriorates the price efficiency of the market. These findings are consistent with the view that more uninformed or speculative options trading can increase the volatility of the underlying assets and deteriorate market efficiency by transmitting additional noise in the market.

Our analysis provides supportive evidence that retail investors play a key role in driving 0DTE options trading activity. By leveraging Robinhood outages as exogenous shocks to retail participation, we observe a significant decline in 0DTE trading during outage periods following the introduction of weekly index options. This highlights that retail investors constitute a sizable portion of 0DTE market participants, and 0DTE options trading serves as a channel through which speculative retail activity amplifies volatility.

This paper contributes to the existing literature on options trading and its impact on underlying assets and price efficiency, which has shown mixed findings. We provide evidence that the increase in 0DTE options trading leads to higher volatility of underlying assets and reduced price efficiency. Also, we offer new insight into how retail-driven trading of short-term options can amplify volatility, reinforcing theoretical predictions about the destabilizing effects of noise traders.

Furthermore, this paper contributes to option literature by examining unexplored dimensions of time-to-expiration. By exploring this new perspective, this paper provides evidence that the relationship between options trading and volatility depends on the type of options traded. In this regard, this paper advances the understanding of the impact of short-term options trading on its underlying assets, in contrast to prior studies that concentrate on the effects of general options trading.

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Tables and Figures

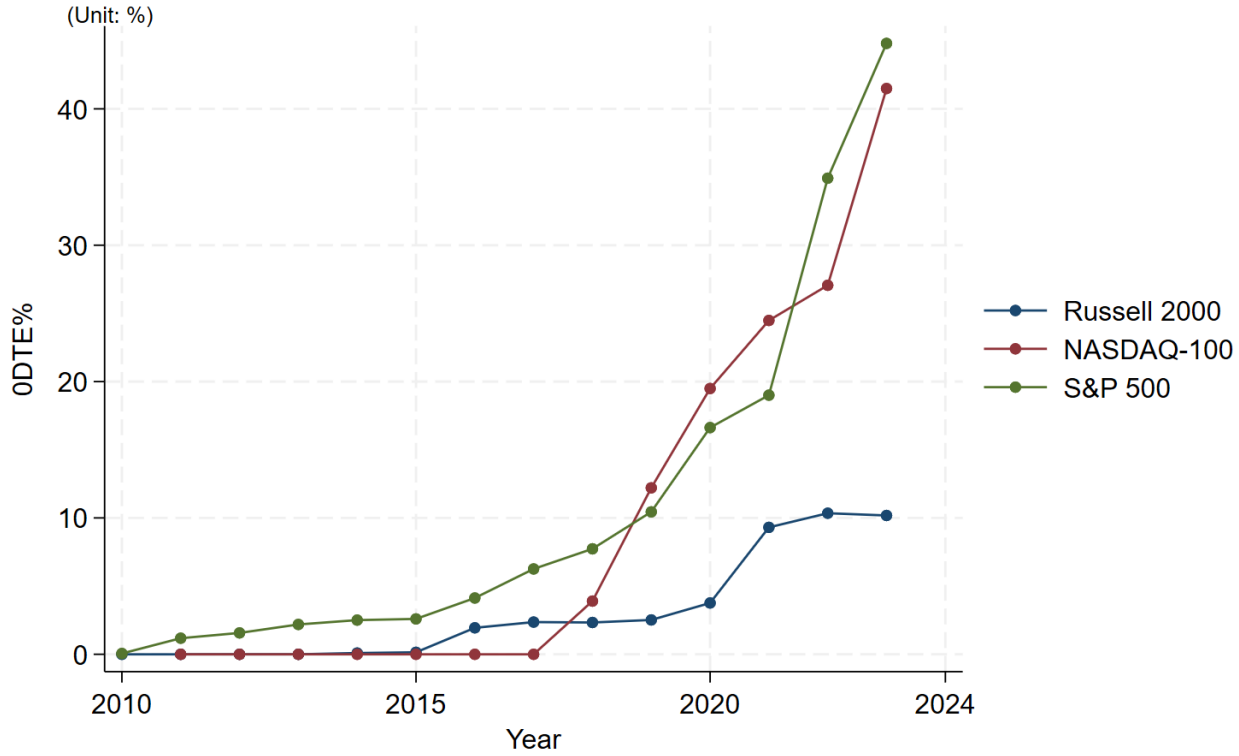


Figure 1: The Trend of Average Daily 0DTE%

This figure shows the average daily trading volume of 0DTE as a percentage of the total trading volume for the index-linked options (as denoted by 0DTE%) from January 2010 to August 2023. The blue line represents the 0DTE% of the Russell 2000, the red line represents the 0DTE% of the NASDAQ-100, and the green line represents the 0DTE% of the S&P 500 index, respectively.

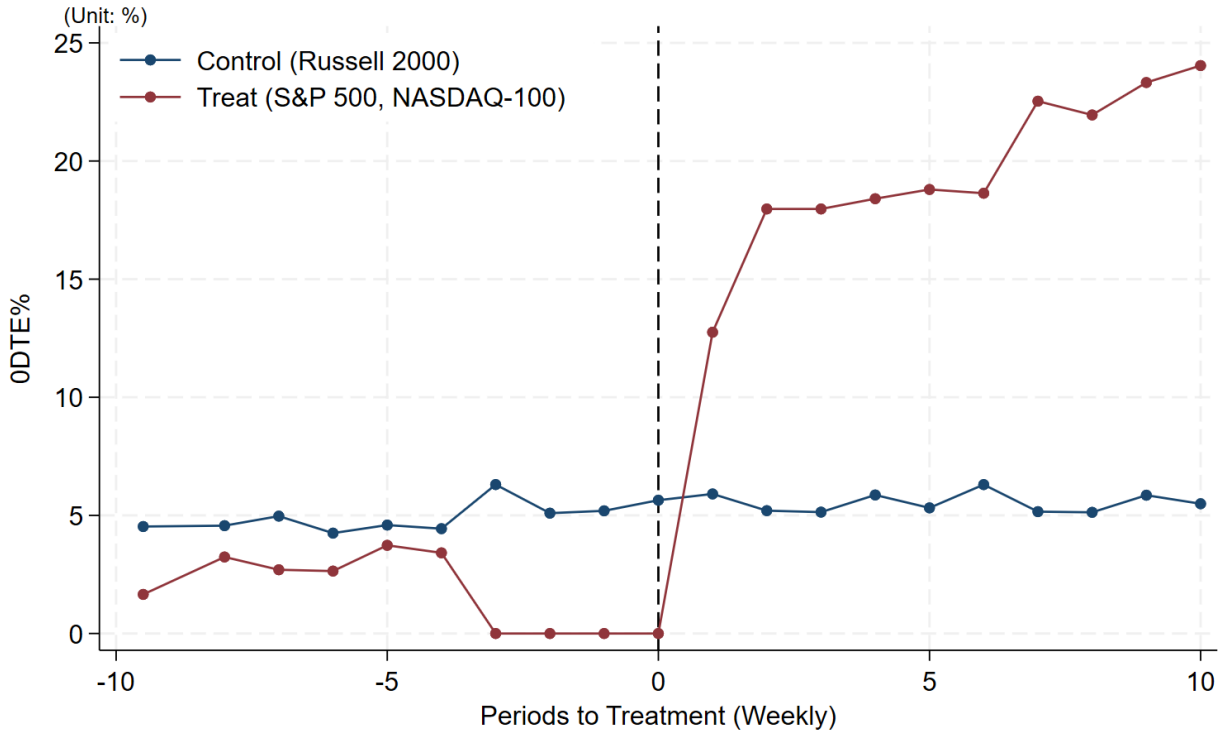


Figure 2: Event Study of 0DTE% Around Weekly Option Introductions

This figure plots event-time averages of 0DTE options trading (0DTE%) by stacking the weekly observations around the introduction before and after a 10-week window. The series is constructed using a stacked event-time framework where each index's treatment period is aligned at event time zero. The blue line represents the control group (Russell 2000), and the red line represents the treated group (S&P 500 and NASDAQ-100). The dashed vertical line denotes the week of the option introduction.

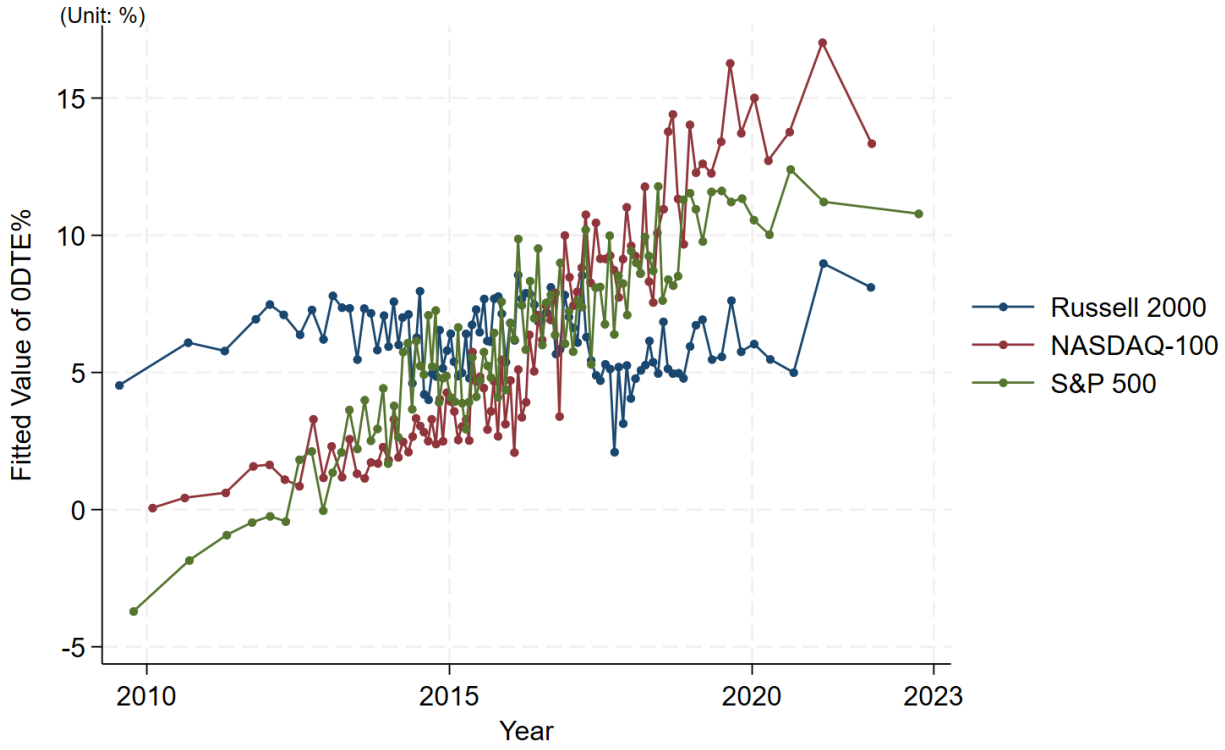


Figure 3: The Trend of Fitted Value of 0DTE%

This figure shows the daily fitted value of 0DTE as a percentage of the total trading volume for the index-linked options (as denoted by 0DTE%) from January 2010 to August 2023. We use the fitted value of 0DTE% from the first stage of IV regression from equation (4), including control variables and the index and day of the week fixed effects. The blue line represents the 0DTE% of the Russell 2000, the red line represents the 0DTE% of the NASDAQ-100, and the green line represents the 0DTE% of the S&P500 index, respectively.

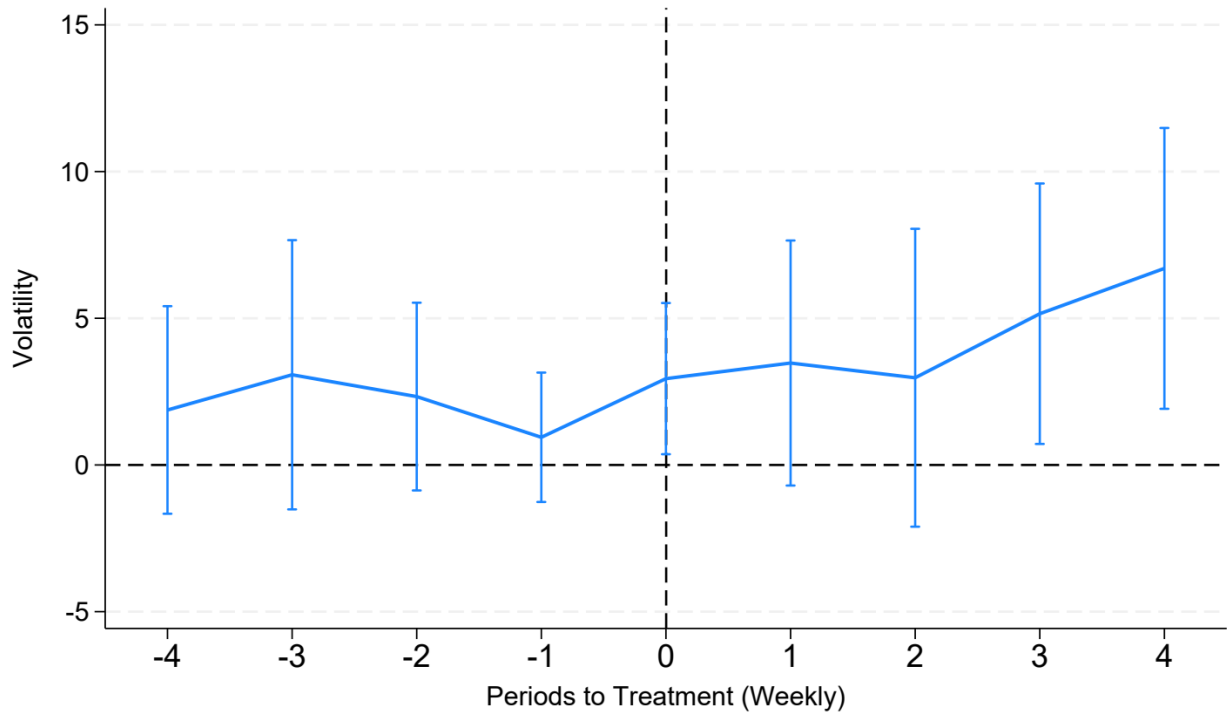


Figure 4: Event Study of Volatility Around Weekly Option Introductions

This figure plots the weekly mean difference in volatility between treated and control indices around the introduction of weekly index options. We align each index to its own introduction date, stack observations within a four-week window before and after the event, and compute the mean difference at each event time. The blue line shows the estimated event-time coefficients, and the vertical bars represent 95 percent confidence intervals.

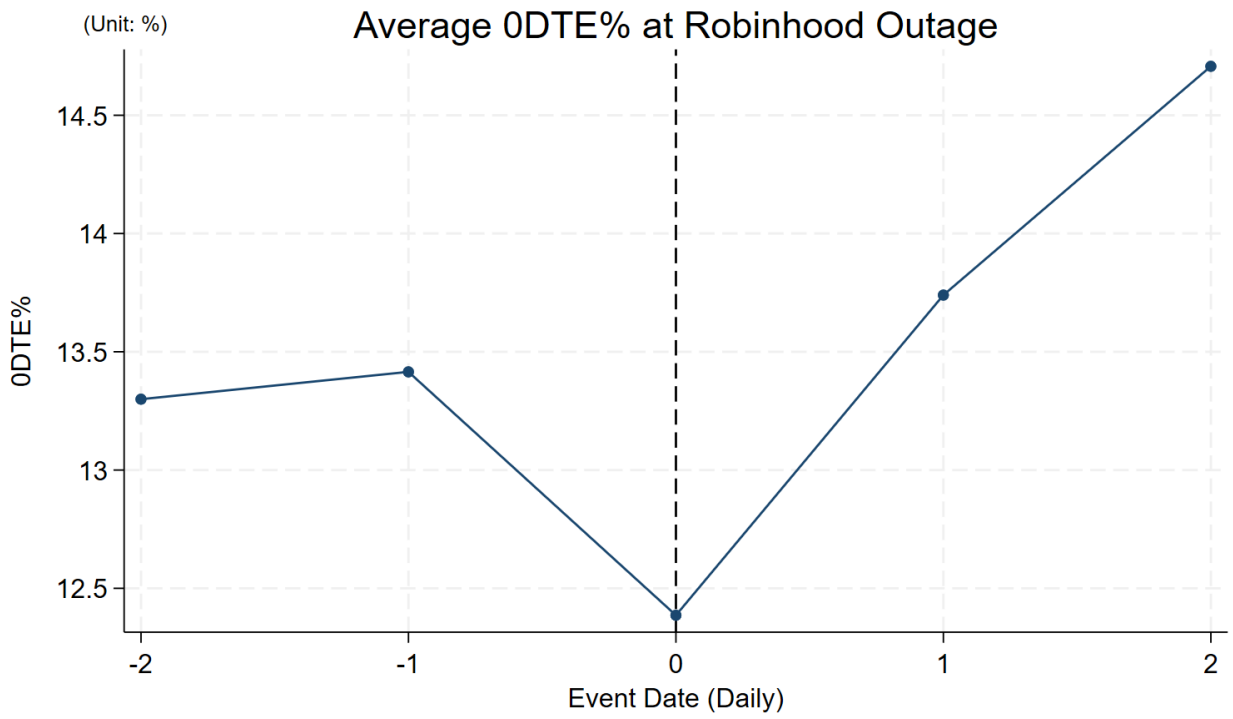


Figure 5: The Average Daily 0DTE% Around Robinhood Outages

This figure presents the average daily trading volume of 0DTE as a percentage of the total trading volume for the index-linked options (as denoted by 0DTE%) around Robinhood outages. The sample includes all identified outages after 2018, because Robinhood has provided options trading since 2018. Day 0 denotes the outage event date.

Table 1: Introduction of Weekly Index Options

The table presents detailed information on the introduction of weekly index options for the S&P 500, NASDAQ-100, and Russell 2000 index. We report the index name, option name, and effective date of introduction. The table displays information on the S&P 500 index in column (1), the NASDAQ-100 index in column (2), and the Russell 2000 index in column (3), respectively. (Source: the Chicago Board Options Exchange (CBOE) and the Chicago Mercantile Exchange (CME))

	(1)	(2)	(3)
Underlying Index:	S&P 500	NASDAQ-100	Russell 2000
Option Name:	SPXW	NDXP	RUTW
Introduction date for:			
Monday	15th August, 2016	21st February, 2020	N/A
Tuesday	18th April, 2022	8th August, 2022	8th January, 2024
Wednesday	23rd February, 2016	19th April, 2018	N/A
Thursday	11th May, 2022	8th August, 2022	8th January, 2024
Friday	2nd December, 2010	4th January, 2018	N/A

Table 2: Descriptive Statistics

The table presents descriptive statistics for our main dependent variable ($\sigma_{i,t}$), independent variable ($0DTE\%_{i,t}$), and control variables. Details of the variables are stated in Table IA1. Panel A reports descriptive statistics for the variables, and Panel B reports the correlations between the variables. The sample covers from January 2010 to August 2023.

Panel A: Summary Statistics

	(1) N	(2) Mean	(3) Std. Dev.	(4) Min	(5) Median	(6) Max
$\sigma_{i,t}$ (bp)	10,010	9.56	5.47	2.22	8.22	42.22
$0DTE\%_{i,t}$ (%)	10,010	7.37	13.81	0.00	0.00	65.12
$Index\ Return_{i,t}$ (%)	10,010	0.05	1.28	-13.27	0.08	9.15
$Index\ Volume_{i,t}$ (1 million)	10,010	65.54	57.67	4.33	47.56	709.32
$\sigma_{i,t-5}$ (bp)	10,005	9.55	5.48	2.22	8.22	42.22
$CP\ Spread_t$	9,119	0.17	0.14	-0.37	0.13	2.57
$Default\ Spread_t$	9,931	2.48	0.48	1.56	2.38	4.31
EPU_t	10,010	1.22	0.85	0.03	1.01	8.08
σ_t^{forex}	9,858	0.31	0.11	0.10	0.29	0.87
σ_t^{cpi}	10,010	0.70	0.47	0.19	0.53	1.94

Panel B: Correlations

	(1) $\sigma_{i,t}$	(2) $0DTE\%_{i,t}$	(3) $Index\ Return_{i,t}$	(4) $Index\ Volume_{i,t}$	(5) $\sigma_{i,t-5}$	(6) $CP\ Spread_t$	(7) $Default\ Spread_t$	(8) EPU_t	(9) σ_t^{forex}	(10) σ_t^{cpi}
(1) $\sigma_{i,t}$	1.00									
(2) $0DTE\%_{i,t}$	0.09	1.00								
(3) $Index\ Return_{i,t}$	-0.18	0.01	1.00							
(4) $Index\ Volume_{i,t}$	0.24	0.00	-0.10	1.00						
(5) $\sigma_{i,t-5}$	0.65	0.09	0.03	0.06	1.00					
(6) $CP\ Spread_t$	0.21	-0.08	0.02	0.05	0.22	1.00				
(7) $Default\ Spread_t$	0.17	-0.33	0.02	0.26	0.20	0.11	1.00			
(8) EPU_t	0.36	0.15	0.06	0.12	0.38	0.03	0.25	1.00		
(9) σ_t^{forex}	0.31	0.05	0.02	0.17	0.36	0.32	0.39	0.26	1.00	
(10) σ_t^{cpi}	0.20	0.34	-0.01	0.15	0.20	-0.14	-0.16	0.13	0.18	1.00

Table 3: What Drives 0DTE Options Trading

The table presents evidence about the association between 0DTE% and control variables to identify the main drivers of 0DTE options trading. The regression is

$$0DTE\%_{i,t} = \alpha + \beta' X_{i,t} + \omega_i + \gamma_t + \varepsilon_{i,t},$$

where $0DTE\%_{i,t}$ is computed as the ratio of the total volume of 0DTE options traded over the total trading volume of the index-linked options expressed as a percentage, $X_{i,t}$ is a vector of control variables, ω_i is an index fixed effect, and γ_t is a day of the week fixed effect. Details of the control variables are stated in Table IA1. The *t*-statistics are presented in parentheses. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively. In column (4), the standard errors are clustered at the daily level. The sample covers the period from January 2010 to August 2023.

Dependent variable =	$0DTE\%_{i,t}$			
	(1)	(2)	(3)	(4)
$Index\ Return_{i,t}$	-0.065	-0.252***	-0.214**	-0.214*
	(-0.65)	(-2.76)	(-2.47)	(-1.83)
$Index\ Volume_{i,t}$	-0.005**	-0.044***	-0.047***	-0.047***
	(-2.22)	(-15.75)	(-17.52)	(-17.42)
$\sigma_{i,t-5}$	0.079***	0.336***	0.358***	0.358***
	(2.94)	(12.85)	(14.39)	(9.42)
$CP\ Spread_t$	-0.805	-5.122***	-5.206***	-5.206***
	(-0.85)	(-5.92)	(-6.34)	(-3.99)
$Default\ Spread_t$	-3.087***	-9.012***	-8.964***	-8.964***
	(-17.39)	(-32.12)	(-33.70)	(-26.31)
EPU_t	2.619***	2.272***	2.156***	2.156***
	(14.14)	(13.55)	(13.45)	(9.02)
σ_t^{forex}	13.117***	10.703***	10.918***	10.918***
	(9.19)	(8.29)	(8.92)	(6.17)
σ_t^{cpi}	8.986***	6.655***	6.578***	6.578***
	(30.38)	(23.52)	(24.52)	(13.17)
Observations	9,047	9,047	9,047	9,047
Adjusted R-squared	0.306	0.264	0.286	0.286
Index FE	No	Yes	Yes	Yes
Days of week FE	No	No	Yes	Yes
Clustered Standard Errors	No	No	No	Daily

Table 4: OLS Analysis of 0DTE Options Trading and Volatility

The table presents the association between 0DTE options trading and volatility. The regression is

$$\sigma_{i,t} = \alpha + \beta\ 0DTE\%_{i,t} + \delta' X_{i,t} + \omega_i + \gamma_t + \varepsilon_{i,t},$$

where $\sigma_{i,t}$ is the daily standard deviation of the index's midpoint quote return every 5 minutes, $0DTE\%_{i,t}$ is computed as the ratio of the total volume of 0DTE options traded over the total trading volume of the index-linked options expressed as a percentage, $X_{i,t}$ is a vector of control variables, ω_i is an index fixed effect, and γ_t is a day of the week fixed effect. Details of the control variables are stated in Table IA1. The *t*-statistics are presented in parentheses. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively. The standard errors are clustered at the daily level. The sample covers the period from January 2010 to August 2023.

Dependent variable =	$\sigma_{i,t}$		
	(1)	(2)	(3)
$0DTE\%_{i,t}$	0.068***	0.060***	0.056***
	(12.54)	(14.35)	(11.15)
$Index\ Return_{i,t}$		-0.615***	-0.614***
		(-7.78)	(-8.27)
$Index\ Volume_{i,t}$		0.047***	0.047***
		(28.42)	(25.65)
$\sigma_{i,t-5}$		0.483***	0.399***
		(26.04)	(22.23)
$CP\ Spread_t$			3.854***
			(7.37)
$Default\ Spread_t$			-0.555***
			(-3.61)
EPU_t			0.828***
			(7.99)
σ_t^{forex}			0.951
			(1.54)
σ_t^{cpi}			-0.000
			(-0.00)
Observations	10,010	10,005	9,047
Adjusted R-squared	0.027	0.568	0.581
Index FE	Yes	Yes	Yes
Days of week FE	Yes	Yes	Yes

Table 5: IV Analysis of 0DTE Options Trading and Volatility

The table presents a causal impact of 0DTE options trading and volatility using IV regression. The first-stage regression is

$$0DTE\%_{i,t} = \alpha + \beta\ introduction_{i,t} + \delta' X_{i,t} + \omega_i + \gamma_t + u_{i,t},$$

where $0DTE\%_{i,t}$ is computed as the ratio of the total volume of 0DTE options traded over the total trading volume of the index-linked options expressed as a percentage, $introduction_{i,t}$ is the dummy set to zero before the introduction of the index weekly option and one afterward, $X_{i,t}$ is a vector of control variables, ω_i is an index fixed effect, and γ_t is a day of the week fixed effect. The second-stage regression is

$$\sigma_{i,t} = \alpha + \beta\ 0\widehat{DTE\%}_{\iota,t} + \delta' X_{i,t} + \omega_i + \gamma_t + \varepsilon_{i,t},$$

where $\sigma_{i,t}$ is the daily standard deviation of the index's midpoint quote return every 5 minutes, $0\widehat{DTE\%}_{\iota,t}$ is the estimated 0DTE% from the first-stage regression, $X_{i,t}$ is a vector of control variables, ω_i is an index fixed effect, and γ_t is a day of the week fixed effect. Columns (1) to (3) report the estimates from the first stage of IV regression, and columns (4) to (6) report the estimates from the second stage of IV regression, where the dependent variable is the σ_{it}, with Kleibergen-Paap *F*-statistic following Kleibergen and Paap (2006). Details of the control variables are stated in Table IA1. The *t*-statistics are presented in parentheses. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively. The standard errors are clustered at the daily level. The sample covers the period from January 2010 to August 2023.

Dependent variable =	First-stage estimates $0DTE\%_{i,t}$			Second-stage estimates $\sigma_{i,t}$		
	(1)	(2)	(3)	(4)	(5)	(6)
$Introduction_{i,t}$	27.268***	26.406***	21.946***			
	(71.92)	(67.90)	(59.25)			
$0\widehat{DTE\%}_{\iota,t}$				0.069***	0.068***	0.063***
				(11.42)	(15.24)	(11.62)
$Index\ Return_{i,t}$		-0.090	-0.099		-0.613***	-0.613***
		(-0.95)	(-1.21)		(-7.75)	(-8.27)
$Index\ Volume_{i,t}$		-0.022***	-0.024***		0.048***	0.047***
		(-9.93)	(-12.15)		(28.38)	(25.66)
$\sigma_{i,t-5}$		0.238***	0.158***		0.479***	0.397***
		(9.19)	(5.96)		(25.80)	(22.05)
$CP\ Spread_t$			-4.985***			3.892***
			(-4.98)			(7.49)
$Default\ Spread_t$			-3.492***			-0.490***
			(-15.02)			(-3.16)
EPU_t			0.993***			0.812***
			(5.77)			(7.87)
σ_t^{forex}			4.806***			0.871
			(3.93)			(1.41)
σ_t^{cpi}			5.690***			-0.049
			(15.99)			(-0.31)
Observations	10,010	10,005	9,047	10,010	10,005	9,047
Adjusted R-squared				0.027	0.567	0.581
Index FE	Yes	Yes	Yes	Yes	Yes	Yes
Days of week FE	Yes	Yes	Yes	Yes	Yes	Yes
Kleibergen-Paap *F*-statistic	5,172	4,611	3,511			

Table 6: Price Efficiency: Hasbrouck's Pricing Error Test

The table reports the impact of 0DTE options trading on price efficiency using Hasbrouck's (1993) pricing error, which is indicated as $\sigma_{i,t}^{HB}$. The regression is

$$Y = \alpha + \beta 0\widehat{DTE\%}_{i,t} + \delta' X_{i,t} + \omega_i + \gamma_t + \varepsilon_{i,t},$$

where Y takes one of our main dependent variables: Hasbrouck's pricing error at t or t+1, $\sigma_{i,t}^{HB}$ or $\sigma_{i,t+1}^{HB}$, respectively. $\sigma_{i,t}^{HB}$ and $\sigma_{i,t+1}^{HB}$ are the daily standard deviation of the pricing error following Hasbrouck (1993), $0\widehat{DTE\%}_{i,t}$ is the estimated 0DTE% from the first-stage regression in equation (4). $X_{i,t}$ is a vector of control variables, ω_i is an index fixed effect, and γ_t is a day of the week fixed effect. Details of the control variables are stated in Table IA1. Columns (1) and (2) display the contemporaneous model, while columns (3) and (4) report the lagged model. The *t*-statistics are presented in parentheses. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively. We report Kleibergen-Paap *F*-statistic following Kleibergen and Paap (2006). The standard errors are clustered at the daily level. The sample covers the period from January 2010 to August 2023.

Dependent variable =	$\sigma_{i,t}^{HB}$		$\sigma_{i,t+1}^{HB}$	
	(1)	(2)	(3)	(4)
$0\widehat{DTE\%}_{i,t}$	0.246***	0.241***	0.222***	0.219***
	(13.12)	(9.07)	(11.29)	(7.97)
$Index\ Return_{i,t}$	0.004	0.008	-0.060**	-0.057**
	(0.12)	(0.27)	(-2.54)	(-2.50)
$Index\ Volume_{i,t}$	0.245***	0.285***	0.048***	0.075***
	(12.31)	(13.03)	(2.64)	(3.85)
$\sigma_{i,t-5}$	0.243***	0.154***	0.316***	0.242***
	(9.74)	(6.36)	(12.46)	(9.37)
$CP\ Spread_t$		0.167***		0.181***
		(5.59)		(7.73)
$Default\ Spread_t$		-0.140***		-0.095***
		(-6.46)		(-4.34)
EPU_t		0.094***		0.091***
		(4.88)		(5.02)
σ_t^{forex}		0.049**		0.035*
		(2.57)		(1.89)
σ_t^{cpi}		-0.121***		-0.066***
		(-6.28)		(-3.29)
Observations	9,753	8,803	9,754	8,803
Adjusted R-squared	0.161	0.181	0.136	0.149
Index FE	Yes	Yes	Yes	Yes
Days of week FE	Yes	Yes	Yes	Yes
Kleibergen-Paap *F*-statistic	3,762	2,343	3,763	2,343

Table 7: Do Retail Investors Drive the 0DTEs?

The table presents the association between 0DTE options trading and volatility. The regression is

$$0DTE\%_{i,t} = \alpha + \beta_1\, Outage_t + \delta' X_{i,t} + \omega_i + \gamma_t + \varepsilon_{i,t},$$

where $0DTE\%_{i,t}$ is computed as the ratio of the total volume of 0DTE options traded over the total trading volume of the index-linked options expressed as a percentage, $Outage_t$ is dummy set to one if Robinhood has an outage during trading hours and zero otherwise, $X_{i,t}$ is a vector of control variables, ω_i is an index fixed effect, and γ_t is a day of the week fixed effect. Details of the control variables are stated in Table IA1. The *t*-statistics are presented in parentheses. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively. The standard errors are clustered at the daily level. The sample in this table covers the period from January 2018 to August 2023, since Robinhood provides options trading after 2018.

Dependent variable =	$0DTE\%_{i,t}$				
	All Outages	Outages ≥ 30 mins	Outages ≥ 60 mins	Outages ≥ 120 mins	Outages < 30 mins
	(1)	(2)	(3)	(4)	(4)
$Outage_t$	-2.552**	-4.534***	-5.903***	-4.311**	0.760
	(-2.12)	(-2.94)	(-3.86)	(-2.18)	(0.51)
$Index\ Return_{i,t}$	-0.120	-0.137	-0.144	-0.130	-0.121
	(-0.76)	(-0.87)	(-0.91)	(-0.83)	(-0.76)
$Index\ Volume_{i,t}$	-0.028***	-0.028***	-0.028***	-0.028***	-0.028***
	(-3.61)	(-3.62)	(-3.63)	(-3.66)	(-3.69)
$\sigma_{i,t-5}$	-0.058	-0.056	-0.055	-0.056	-0.059
	(-1.02)	(-0.98)	(-0.96)	(-0.99)	(-1.03)
$CP\ Spread_t$	-8.561***	-8.417***	-8.383***	-8.474***	-8.499***
	(-6.51)	(-6.40)	(-6.37)	(-6.44)	(-6.42)
$Default\ Spread_t$	2.899**	2.836**	2.899**	2.750**	2.612**
	(2.55)	(2.51)	(2.57)	(2.43)	(2.30)
EPU_t	-0.373	-0.376	-0.384	-0.379	-0.370
	(-0.82)	(-0.83)	(-0.85)	(-0.84)	(-0.82)
σ_t^{forex}	40.697***	40.662***	40.744***	40.742***	41.026***
	(10.67)	(10.67)	(10.72)	(10.67)	(10.72)
σ_t^{cpi}	9.615***	9.635***	9.652***	9.638***	9.643***
	(16.01)	(16.07)	(16.12)	(16.05)	(16.01)
Constant	-6.625***	-6.510***	-6.665***	-6.378***	-6.213***
	(-2.99)	(-2.95)	(-3.02)	(-2.88)	(-2.80)
Observations	3,456	3,456	3,456	3,456	3,456
Adjusted R-squared	0.485	0.486	0.487	0.485	0.485
Index FE	Yes	Yes	Yes	Yes	Yes
Days of week FE	Yes	Yes	Yes	Yes	Yes

Table 8: Robustness: Subsample Analysis

The table presents a causal impact of 0DTE options trading and volatility, using alternative subsample periods. The regression is

$$\sigma_{i,t} = \alpha + \beta\ 0\widehat{DTE\%}_{i,t} + \delta' X_{i,t} + \omega_i + \gamma_t + \varepsilon_{i,t},$$

where $\sigma_{i,t}$ is the daily standard deviation of the index's midpoint quote return every 5 minutes, $0\widehat{DTE\%}_{i,t}$ is the estimated 0DTE% from the first-stage regression in equation (4), $X_{i,t}$ is a vector of control variables, ω_i is an index fixed effect, and γ_t is a day of the week fixed effect. Columns (1) to (5) report the estimates using event windows of 1, 2, 3, 6, and 12 months before and after the introduction of weekly options in each index, respectively. Details of the control variables are stated in Table IA1. The *t*-statistics are presented in parentheses. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively. We report Kleibergen-Paap *F*-statistic following Kleibergen and Paap (2006). The standard errors are clustered at the daily level.

Dependent variable =	$\sigma_{i,t}$				
Subsample Period =	*1-month window*	*2-month window*	*3-month window*	*6-month window*	*12-month window*
	(1)	(2)	(3)	(4)	(5)
$0\widehat{DTE\%}_{i,t}$	0.062***	0.059***	0.076***	0.104***	0.076***
	(3.53)	(3.97)	(6.25)	(11.92)	(11.18)
$Index\ Return_{i,t}$	-0.622***	-0.586***	-0.580***	-0.545***	-0.590***
	(-3.65)	(-4.11)	(-4.61)	(-5.29)	(-6.56)
$Index\ Volume_{i,t}$	0.060***	0.069***	0.065***	0.054***	0.046***
	(9.87)	(10.76)	(12.16)	(15.12)	(20.92)
$\sigma_{i,t-5}$	0.443***	0.363***	0.371***	0.358***	0.402***
	(10.13)	(10.24)	(11.29)	(14.48)	(18.54)
$CP\ Spread_t$	2.634**	2.872***	2.546***	2.101***	2.817***
	(1.98)	(3.16)	(3.03)	(3.01)	(4.27)
$Default\ Spread_t$	0.083	-0.534	-0.611**	-0.183	-0.114
	(0.24)	(-1.50)	(-1.98)	(-0.77)	(-0.56)
EPU_t	-0.881***	-0.766***	-0.335	-0.016	0.458***
	(-2.75)	(-3.37)	(-1.50)	(-0.10)	(3.00)
σ_t^{forex}	-6.260***	-1.687	0.021	0.680	0.524
	(-2.86)	(-1.13)	(0.01)	(0.71)	(0.67)
σ_t^{cpi}	2.763***	2.907***	2.245***	1.043***	0.040
	(4.86)	(5.22)	(4.57)	(4.07)	(0.22)
Observations	821	1,505	2,041	3,649	5,567
Adjusted R-squared	0.648	0.624	0.627	0.593	0.569
Index FE	Yes	Yes	Yes	Yes	Yes
Days of week FE	Yes	Yes	Yes	Yes	Yes
Kleibergen-Paap *F*-statistic	193.6	356.8	510.7	1059	2068

Table 9: Robustness: Spillover Effects

The table presents a causal impact of 0DTE options trading on several spillover measures. The regression is

$$Y_{i,t} = \alpha + \beta\, 0\widehat{DTE\%}_{i,t} + \delta' X_{i,t} + \omega_i + \gamma_t + \varepsilon_{i,t},$$

where $Y_{i,t}$ takes one of our main dependent variables: absolute and squared returns for close-to-close ($|Ret_{i,t}^{CC}|$ or $\left(Ret_{i,t}^{CC}\right)^2$), open-to-close ($|Ret_{i,t}^{OC}|$ or $\left(Ret_{i,t}^{OC}\right)^2$), or close-to-close ($|Ret_{i,t}^{Overnight}|$ or $\left(Ret_{i,t}^{Overnight}\right)^2$) returns, respectively. $0\widehat{DTE\%}_{i,t}$ is the estimated 0DTE% from the first-stage regression in equation (4), $X_{i,t}$ is a vector of control variables, ω_i is an index fixed effect, and γ_t is a day of the week fixed effect. Details of the control variables are stated in Table IA1. The *t*-statistics are presented in parentheses. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively. We report Kleibergen-Paap *F*-statistic following Kleibergen and Paap (2006). The standard errors are clustered at the daily level. The sample covers the period from January 2010 to August 2023.

Dependent variable=	$\|Ret_{i,t}^{CC}\|$	$\|Ret_{i,t}^{OC}\|$	$\|Ret_{i,t}^{Overnight}\|$	$\left(Ret_{i,t}^{CC}\right)^2$	$\left(Ret_{i,t}^{OC}\right)^2$	$\left(Ret_{i,t}^{Overnight}\right)^2$
	(1)	(2)	(3)	(4)	(5)	(6)
$0\widehat{DTE\%}_{i,t}$	0.0056***	0.0049***	0.0018**	0.0002***	0.0001***	0.0000
	(4.29)	(5.09)	(2.18)	(2.93)	(3.86)	(1.02)
$Index\ Return_{i,t}$	0.0071***	0.0052***	0.0023***	0.0003***	0.0001***	0.0001***
	(17.05)	(17.24)	(7.18)	(8.71)	(10.65)	(3.08)
$Index\ Volume_{i,t}$	-0.0259	-0.0329**	-0.0385**	-0.0044	-0.0012*	-0.0007
	(-0.73)	(-2.30)	(-2.01)	(-1.10)	(-1.81)	(-0.42)
$\sigma_{i,t-5}$	0.0432***	0.0248***	0.0304***	0.0022***	0.0007***	0.0011***
	(8.83)	(9.40)	(8.06)	(4.48)	(6.91)	(4.14)
$CP\ Spread_t$	-0.0444	-0.0655**	0.0693***	-0.0000	-0.0022**	0.0023**
	(-1.33)	(-2.43)	(3.07)	(-0.00)	(-2.46)	(2.49)
$Default\ Spread_t$	-0.0665***	-0.0360***	-0.0508***	-0.0025***	-0.0008***	-0.0017***
	(-5.09)	(-3.68)	(-5.84)	(-3.72)	(-3.00)	(-4.26)
EPU_t	0.0978***	0.0565***	0.0801***	0.0063***	0.0024***	0.0027***
	(4.28)	(3.62)	(5.12)	(3.84)	(3.55)	(3.54)
σ_t^{forex}	0.4796***	0.1925*	0.1336	0.0220**	0.0073**	0.0033
	(3.24)	(1.84)	(1.22)	(2.54)	(2.08)	(0.67)
σ_t^{cpi}	-0.1274***	-0.0151	-0.0620**	-0.0109***	-0.0018*	-0.0048***
	(-3.51)	(-0.58)	(-2.39)	(-3.78)	(-1.80)	(-2.73)
Observations	9,853	9,853	9,850	9,853	9,853	9,850
Adjusted R-squared	0.2719	0.2222	0.2045	0.1851	0.1641	0.0983
Index FEs	Yes	Yes	Yes	Yes	Yes	Yes
Day-of-the-week FEs	Yes	Yes	Yes	Yes	Yes	Yes
Kleibergen-Paap F-statistic	4,060	4,060	4,056	4,060	4,060	4,056

Internet Appendix

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Does 0DTE Options Trading Increase Volatility?

IA1. Discussion

The overall impact of 0DTE options trading on market volatility remains a subject of ongoing debate. In this section, we discuss other papers that have proposed the opposite effect from our results. We first review these studies, which generally document a decline in volatility affected by 0DTE options trading. Furthermore, to assess the overall impact, we provide additional analyses demonstrating that 0DTE trading drives an overall increase in volatility, even after accounting for the alternative mechanisms proposed by this prior research. Finally, we then highlight the key differences in empirical design that may generate these conflicting results.

IA1.1. Divergent Perspectives on Options and Volatility: Market Makers' Net Gamma Channel

Dim et al. (2023) and Adam et al. (2024) emphasize a market maker channel in which 0DTE options trading primarily affects volatility through dealers' net gamma exposure. In these studies, high 0DTE activity is typically associated with positive net gamma for option market makers, so that their hedging trades decrease volatility. Similarly, Vasquez et al. (2025) are broadly consistent with this view in that dealers' gamma exposure is usually positive, which tends to attenuate volatility rather than amplify it.

We agree that these findings are highly relevant and informative about the dealer-hedging side of the market, and we do not view them as inconsistent with our results. The existing papers are designed explicitly to isolate the market-maker net gamma channel. Their empirical designs treat the availability or expansion of 0DTE contracts as a way to create variation in dealers' net gamma and then ask whether this induced change in net gamma stabilizes or destabilizes volatility. By design, their approach focuses only on the net gamma channel and leaves out all other effects of 0DTE trading on volatility. In other words, these studies estimate the net-gamma hedging component of the overall 0DTE impact on volatility, not the total effect of 0DTE

trading itself. By contrast, our study is designed to quantify the overall effect of 0DTE options trading on volatility, with particular attention to the retail-investor channel. We focus on speculative 0DTE trading by retail investors that transmit the noise into the price, while allowing for the possibility that market-maker hedging partly offsets this initial shock.

To make this mechanism concrete, consider a hypothetical example. Suppose a surge in speculative 0DTE options trading initially causes market volatility to increase by 10%. In response to these price movements, market makers are compelled to execute gamma hedging. This hedging activity, which involves trading the underlying asset against the prevailing trend, acts as a stabilizing force and might reduce volatility by 3%. The net result is about 7% increase in volatility, illustrating how the hedging effect only partially offsets the initial shock.

It should be emphasized that the dampening effect from hedging is inherently unlikely to exceed the initial volatility increase. This is because hedging is fundamentally a reactive mechanism. Market makers' objective is not to bet against the market trend but to neutralize the delta risk they face from the initial order flow of speculative trading. Their actions provide liquidity and act as a brake on the initial impulse, but they do not constitute a countervailing force powerful enough to fully erase the impact of the primary speculative flow. Therefore, while hedging can mitigate volatility, it cannot fully prevent the net increase caused by significant 0DTE trading activity.

To further substantiate our argument, we conduct an additional robustness analysis controlling the market-maker hedging channel. Following Baltussen et al. (2021), we construct the daily net gamma exposure of option market makers for each index ($\Gamma_{i,t}^{Net}$) and include it as an additional control variable in our baseline specification in equation (5). Furthermore, to ensure that our estimates are not fully dampened by net gamma exposure, we also implement orthogonalized specifications in which both the 0DTE options trading and realized volatility are adjusted to remove variation explained by net gamma exposure.

First, we orthogonalize 0DTE options trading with aggregate market makers' gamma exposure. The estimated equation is:

$$0DTE\%_{i,t} = \alpha + \beta\Gamma_{i,t}^{Net} + \delta' X_{i,t} + \omega_i + \gamma_t + 0DTE\%_{i,t}^{\perp\Gamma^{Net}}, \quad \text{(IA.1)}$$

where $0DTE\%_{i,t}$ is the proportion of 0DTE options volume traded on a given day relative to the total number of index-linked options traded on a given day, $\Gamma_{i,t}^{Net}$ is the net gamma exposure of option market makers for each index, $X_{i,t}$ is a vector of control variables, ω_i is an index fixed effect, and γ_t is a day of the week fixed effect. The standard errors are clustered at the daily level.

From this regression, we calculate the residual, $0DTE\%_{i,t}^{\perp\Gamma^{Net}}$. Then, we estimate the IV approach by replacing $0DTE\%_{i,t}$ with $0DTE\%_{i,t}^{\perp\Gamma^{Net}}$ in equation (4) (i.e., the first stage of IV). In this way, we can remove the part of 0DTE% that comoves with net gamma exposure and examine whether the remaining component of 0DTE% continues to have a significant relationship with volatility.

Next, we orthogonalize volatility with aggregate market makers' gamma exposure to remove any variation of net gamma exposure that correlates with volatility in a similar manner as above.

The estimated equation is:

$$\sigma_{i,t} = \alpha + \beta\Gamma_{i,t}^{Net} + \delta' X_{i,t} + \omega_i + \gamma_t + \sigma_{i,t}^{\perp\Gamma^{Net}}, \quad \text{(IA.2)}$$

where $\sigma_{i,t}$ is the daily standard deviation of the index's midpoint quote return every 5 minutes, $\Gamma_{i,t}^{Net}$ is the net gamma exposure of option market makers for each index, $X_{i,t}$ is a vector of control variables, ω_i is an index fixed effect, and γ_t is a day of the week fixed effect. The standard errors are clustered at the daily level.

By using this equation, we calculate the residual, $\sigma_{i,t}^{\perp\Gamma^{Net}}$. In the same manner, we estimate the IV approach by replacing $\sigma_{i,t}$ with $\sigma_{i,t}^{\perp\Gamma^{Net}}$ in equation (5) to orthogonalize the component of volatility that covaries with net gamma exposure, enabling us to verify whether the remaining variation maintains a significant relationship with 0DTE%.

The results are presented in Table IA2. Columns (1) and (2) report the baseline two-stage IV specification, augmented with the net gamma exposure variable. Columns (3) and (4) use an orthogonalized 0DTE variable ($0DTE\%_{i,t}^{\perp\Delta\Gamma_t}$) which removes any variation from net gamma exposure following equation (11), and columns (5) and (6) use orthogonalized realized volatility ($\sigma_{i,t}^{\perp\Gamma^{Net}}$) which removes any variation from net gamma exposure following equation (12). Across all specifications, the coefficient on $0\widehat{DTE}\%$ remains positive and highly significant in the second stage in columns (2), (4), and (6). This indicates that the relationship between 0DTE options trading and volatility persists even after explicitly controlling gamma-related hedging demand. Specifically, column (4) documents that 0DTE options trading orthogonal to net gamma leads to higher volatility, while column (6) demonstrates that increased 0DTE options trading raises the component of market volatility that is independent of net gamma exposure. In other words, regardless of whether we account for a market maker's net gamma, our findings consistently show that 0DTE options trading increases volatility.

Combined with the prior literature, these results support a two-channel view of 0DTE options trading: a stabilizing hedging channel operating through net gamma, and a destabilizing speculative channel operating through retail-driven trading. Our orthogonalization in columns (3) to (6) shows that the latter channel remains economically and statistically important, rather than being fully absorbed by the former. Overall, our analysis demonstrates that the destabilizing speculative channel outweighs the stabilizing hedging channel, suggesting a net positive overall effect of 0DTE options trading on volatility.

[Insert Table IA2 here]

IA1.2. Mapping 0DTE Options Flow: Volume, Imbalance, and Net Gamma

A specification that conditions only on 0DTE options volume can raise concern. This specification may assume a fixed mapping between volume and order imbalance and constant gamma exposure of dealers. However, since trading volume does not always signal directional pressure, and it varies conditionally with dealers' hedging needs, these factors could be dynamically related to each other. Therefore, we should employ a framework that verifies the joint interactions among volume, order imbalance, and gamma exposure of dealers.

To address this concern, we consider a way to incorporate order imbalance in our specification to complement our results. We augment our volume-based analysis with order-imbalance measures derived from 0DTE put and call activity and examine how their relationship with volatility depends on the sign and magnitude of market makers' net gamma exposure. We re-estimate the OLS and IV specification by including both 0DTE% and 0DTE put–call ratio (i.e., 0DTE PCR defined as $\frac{0DTE\ put}{0DTE\ put+call}$ following Pan and Poteshman (2006)) to proxy the order imbalance of 0DTE options trading. By interacting 0DTE PCR with market makers' net gamma, we can map the impact of order imbalances into dealers' inventory and hedging conditions without requiring trader-type identifiers.

0DTE PCR captures the directional imbalance between put and call trading, so if the coefficient is positive, it suggests that higher relative put volume increases volatility because of more downside hedging demand. Conversely, if it is negative, it implies that higher relative put volume is associated with lower volatility, reflecting volatility compression after heavy downside positioning. However, the market impact of volume or order imbalance is not static. Since market makers' net gamma can dictate their hedging response to order flow, combining it with 0DTE% or 0DTE PCR allows us to capture these dynamics flexibly rather than assuming a fixed relationship.

Accordingly, the sign of the coefficients for both volume (0DTE% × Γ^{Net}) and order imbalance (0DTE PCR × Γ^{Net}) interaction term reflects how dealers' hedging needs moderate the market impact of volume and order imbalance, respectively. If the coefficients are negative, it indicates that when market makers are long gamma, additional 0DTE volume and higher put-call ratios have a dampening impact on volatility, consistent with the stabilizing hedging hypothesis. Conversely, in the short gamma, negative coefficients imply that elevated 0DTE% and 0DTE PCR increase volatility, consistent with destabilizing hedging. On the other hand, if the coefficients are positive, it indicates that when market makers are long gamma, additional 0DTE volume and higher put-call ratios have an amplifying impact on volatility, suggesting that dealer hedging fails to stabilize the speculative flow. Conversely, in the short gamma, it implies that heightened 0DTE% and 0DTE PCR reduce volatility, consistent with the view that dealer hedging attenuates the impact of noisy trading flows. Accordingly, as the volatility impact is conditional on the level of market makers' gamma exposure, we should evaluate the marginal effect of 0DTE options trading at the sample means of net gamma.

Based on this hypothesis, we examine the OLS model by estimating:

$$\sigma_{i,t} = \alpha + \beta_1 0DTE\%_{i,t} + \beta_2 0DTE\ PCR_{i,t} + \beta_3 \Gamma_{i,t}^{Net} + \beta_4 0DTE\ PCR_{i,t} \times \Gamma_{i,t}^{Net} + \beta_5 0DTE\%_{i,t} \times \Gamma_{i,t}^{Net} + \delta' X_{i,t} + \omega_i + \gamma_t + \epsilon_{i,t}, \quad \text{(IA.3)}$$

where $\sigma_{i,t}$ is the daily standard deviation of the index's midpoint quote return every 5 minutes, $0DTE\%_{i,t}$ is the proportion of 0DTE options volume traded on a given day relative to the total number of index-linked options traded on a given day, $0DTE\ PCR_{i,t}$ is the proportion of 0DTE put options volume traded on a given day relative to the sum of put and call options volume traded on a given day, following Pan and Poteshman (2006), $\Gamma_{i,t}^{Net}$ is the net gamma exposure of option market makers for each index, $X_{i,t}$ is a vector

of control variables, ω_i is an index fixed effect, and γ_t is a day of the week fixed effect. The standard errors are clustered at the daily level.

To examine the IV model with these variables, the estimated equation for the first-stage regression is:

$$\sigma_{i,t} = \alpha + \beta_1 0DTE\%_{i,t} + \beta_2 0DTE\ PCR_{i,t} + \beta_3 \Gamma_{i,t}^{Net} + \beta_4 0DTE\ PCR_{i,t} \times \Gamma_{i,t}^{Net} + \beta_5 0DTE\%_{i,t} \times \Gamma_{i,t}^{Net} + \delta' X_{i,t} + \omega_i + \gamma_t + \epsilon_{i,t}, \quad \text{(IA.4)}$$

where $introduction_{i,t}$ is the dummy variable set to zero before the introduction of weekly index options and one afterward, and other variables are computed as described in equation (IA.3).

Then, we regress the second-stage regression by estimating the following equation:

$$\sigma_{i,t} = \alpha + \beta_1 0\widehat{DTE\%}_{\iota,t} + \beta_2 0DTE\ PCR_{i,t} + \beta_3 \Gamma_{i,t}^{Net} + \beta_4 0DTE\ PCR_{i,t} \times \Gamma_{i,t}^{Net} + \beta_5 0DTE\%_{i,t} \times \Gamma_{i,t}^{Net} + \delta' X_{i,t} + \omega_i + \gamma_t + \epsilon_{i,t}, \quad \text{(IA.5)}$$

where $0\widehat{DTE\%}_{\iota,t}$ is the estimated 0DTE% from the first-stage regression, and other variables are computed as described in equation (IA.3).

Table IA3 presents the results for each OLS and IV model, respectively. In columns (1) and (2), the contemporaneous coefficient on 0DTE% remains positive and significant. It implies that 0DTE% increases volatility across columns (1) and (2), so there is a robust volume-driven component to volatility. Specifically, in column (2), we examine the marginal effects of both 0DTE options trading volume and order imbalance evaluated at the sample mean of net gamma at 0.14 (not tabulated). First, for 0DTE options trading volume, the coefficient of 0DTE% is 0.1758, and its interaction with net gamma (0DTE% × Γ^{Net}) is 0.9005. The positive interaction implies that volume-induced volatility is not reduced by gamma hedging. Consequently, the marginal effect at the mean net gamma is approximately 0.30 (i.e., 0.1758 + 0.9005·0.14),

confirming that 0DTE options trading volume continues to exert a net positive pressure on volatility. Second, for 0DTE option order imbalance, the coefficient of 0DTE PCR is 4.6136, while its interaction with net gamma (0DTE PCR × Γ^{Net}) is -0.1525. This negative interaction suggests that market makers' long gamma exposure helps decrease the volatility impact of directional pressure. However, the marginal effect remains positive at approximately 4.59 (i.e., 4.6136 - 0.1525 ·0.14), indicating that hedging activity only partially offsets the destabilizing impact of order imbalances. Therefore, while the gamma hedging channel provides some mitigating offset against order imbalances, the aggregate effect of both 0DTE volume and imbalance results in a robust increase in market volatility.

Overall, by estimating 0DTE volume and imbalance separately and allowing their impact to vary with Γ^{Net}, we relax the constant-order imbalance and constant-gamma assumption and disentangle the market makers' inventory channel from the directional order flow channel. Through this test, we show that the relationship between 0DTE options trading and volatility remains positive and statistically significant even after incorporating order imbalance and market-makers' net gamma exposure. This indicates that 0DTE trading volume continues to exert an incremental impact on volatility, beyond what can be explained by a market maker's inventory and gamma channels. In other words, directional imbalances and dealers' hedging motives are important, but they do not fully explain the overall volatility effects associated with 0DTE options trading. Therefore, these results reinforce our interpretation that 0DTE options trading serves as a vehicle for speculative, non-fundamental trading that amplifies volatility, even though there are also stabilizing effects from market-makers' hedging that other studies emphasize.

[Insert Table IA3 here]

IA1.3. Methodological Differences from Dim et al. (2023)

Dim et al. (2023) study 0DTE options trading and emphasize a market-maker channel in which gamma hedging associated with 0DTE flows can decrease intraday volatility. When 0DTE options trading occurs, dealers are generally long gamma and hedge dynamically, their trading tends to lean against price movements and absorb order-flow imbalances, which shows a lower realized volatility around periods of heavy 0DTE options trading.

In their paper, they argue that they have different results under a similar specification to ours. In their Internet Appendix (IA.1.2), Dim et al. (2023) document that "lagged 0DTE option volume fails to exhibit a statistically significant relationship with contemporaneous realized volatility." However, it should be noted that this specification is explicitly designed to test for next-day spillovers: yesterday's 0DTE options volume is used to explain today's realized volatility. At first glance, this null effect might seem to contradict our findings that 0DTE options trading amplifies volatility. However, our paper's thesis is fundamentally different in scope. We focus on the contemporaneous, same-day impact of 0DTE options trading on volatility, precisely because the distinct feature of 0DTE contracts is their same-day expiration, which should concentrate their price impact within the trading day rather than into the following session.

To clarify this distinction between the within-day impact and spillover effect, we extend our analysis beyond the purely intraday horizon and separate return variation into three horizons: overnight (close-to-open), intraday (open-to-close), and close-to-close returns, following Lou et al. (2019). We construct both absolute and squared returns at each horizon and regress them on our instrumented measure of 0DTE options trading. The results show that the effect of 0DTE options trading is concentrated in the same-day volatility, with little evidence of persistent spillovers into the overnight or next-day close-to-close variation. In this sense, our findings are not in conflict with Dim et al. (2023). In fact, their evidence that lagged 0DTE volume does not predict next-day volatility is not inconsistent with our conclusion that the impact of 0DTE options trading is confined to the same trading day. We discuss and explain more details of this analysis in Table 9 from our 5. Robustness section.

Another difference between theirs and ours is the definition of 0DTE variable. We argue that their variable, which represents 0DTE options trading volume, *DTE0%*, may lead to biased scaling. Their denominator of *DTE0%* is "the corresponding trading volume of all options (with roots SPY, SPX, and SPXW) expiring within the next month." This may mechanically exclude a subset of observations. For example, if an option is issued on October 1 and expires in three months, trading of this option would be excluded from the denominator because its expiration date does not fall within the next month. Given that SPY and SPX options' expiration is usually longer than a month, many of them are likely to be excluded from the computation. In this case, their *DTE0%* overestimates the trading proportion of 0DTE options.

In sum, these differences in horizon and measurement mean that Dim et al. (2023) primarily test whether 0DTE activity has an impact on volatility in the following day, using a *DTE0%* measure that may overstate the share of 0DTE trading in total options volume. In contrast, our design is explicitly targeted at the within-day volatility consequences of 0DTE options trading that considers all available 0DTE options. Accordingly, our evidence of heightened intraday volatility and their finding of limited next-day spillovers are best viewed as complementary, rather than contradictory, characterizations of how 0DTE options trading affects different time profiles of volatility.

IA1.4. Methodological Differences from Adam et al. (2024)

Next, Adam et al. (2024) also argue that 0DTE options trading can attenuate market volatility by stimulating the market maker hedging channel. Their main conclusion is that the introduction and expansion of SPXW 0DTE contracts increase dealers' net gamma exposure, which in turn leads to stabilizing hedging needs and lower realized volatility.

To evaluate their hypothesis, they employ two empirical strategies. First, they use a difference-in-differences (DID) approach, treating days when SPXW options were already available (Monday, Wednesday, and Friday) as the control group and days when SPXW options were not yet available (Tuesday and Thursday) as the treated group. They find that the introduction of Tuesday and Thursday weekly options is associated with a reduction in market volatility. Second, they adopt an IV approach, using the introduction of Tuesday and Thursday SPXW options as an instrument to identify the causal effect of market makers' 0DTE net gamma on market volatility. In this specification, they conclude that the increased gamma hedging needs induced by these introductions reduce volatility.

However, both identification strategies may raise concerns that can bias their estimates. On the DID side, using already-treated days as the control group for newly treated days violates the core assumptions of the DID approach and might generate biased results. The already treated group does not represent a valid counterfactual, because it is itself subject to evolving treatment effects from the earlier SPXW introductions. Its observed trend therefore reflects the impact of ongoing treatment rather than the path that would have prevailed in the absence of any treatment. As rigorously demonstrated by Callaway and Sant'Anna (2021), such a mis-specified DID setup generally recovers a weighted average of heterogeneous and possibly negative treatment effects that does not equal the true average treatment effect and can lead to misleading conclusions about the impact on volatility.

On the IV side, their instrument, the introduction of Tuesday and Thursday SPXW options, is unlikely to affect volatility solely through market makers' gamma hedging, which casts doubt on the exclusion restriction. The arrival of new 0DTE contracts can plausibly influence volatility through multiple channels. In particular, it may not only increase market makers' hedging needs but also stimulate speculative 0DTE trading by retail investors, alter the composition of order flow across maturities, and change the intensity of intraday liquidity provision. If these additional channels affect volatility independently of net gamma,

then the instrument is invalid, and the IV estimates do not have a clean causal interpretation for the gamma hedging channel alone.

In sum, these concerns imply that their DID design may not use a proper untreated counterfactual, and their IV strategy is unlikely to satisfy a condition for a valid instrument, especially the exclusion restriction. However, our paper employs a staggered DID in the first stage with proper treatment and control groups, and our IV strategy satisfies the exclusion restriction. Consequently, our approaches allow for a cleaner and more reliable identification of the causal impact of 0DTE trading on market volatility.

IA2. Additional Robustness and Extension

In this section, we present a comprehensive set of robustness tests to confirm the validity and mechanisms of our main results. We specifically conduct five distinct analyses: first, investigating the effect of 0DTE trading on price efficiency using Variance Ratio (VR) tests; second, comparing the distinct impact of 0DTE options against other short-term expirations (1DTE to 4DTE) using a residual-based horse-race approach; third, assessing the sensitivity of our findings to alternative volatility measures calculated at different frequencies and using noise-robust estimators; fourth, examining the stability of our inference under more stringent econometric specifications, including daily fixed effects and alternative standard error clustering; and finally, addressing potential endogeneity and persistence concerns by employing lagged control variables and multiple volatility lags in both OLS and IV frameworks. Overall, our main findings remain consistent throughout these robustness tests. These results confirm that the positive impact of 0DTE options trading on volatility is robust to alternative methodological choices and effectively mitigates concerns related to microstructure noise, model specification, and potential endogeneity.

IA2.1. Robustness Test for Price Efficiency: Variance Ratio (VR)

In this section, to reinforce our finding about how 0DTE trading affects price efficiency, we employ the VR test widely used in the literature (Lo and MacKinlay 1988; O'Hara and Ye 2011), which is a methodology to measure the degree of price efficiency. The VR test allows us to examine whether the movement of stock prices follows a random walk, implying that stock prices are efficient.

The VR is defined as,

$$VR_{i,t}^{(q)} = \frac{Var(r_{i,t-q\to t})}{Var(r_{i,t}) * q}, \tag{IA.6}$$

where the variance of q-period returns is divided by q times the variance of the single-period returns in the same window for index i, where $i \in$ {S&P 500, NASDAQ-100, Russell 2000}. If the asset prices are informationally efficient and generated by a random walk, then the variance of q-period returns must be q times as large as the variance of single-period returns. In this case, the VR should be equal to 1. For a more intuitive interpretation, we define the absolute value of VR as,

$$AbsVarRatio_{i,t}^{(q)} = \left|VR_{i,t}^{(q)} - 1\right|, \tag{IA.7}$$

where $AbsVarRatio_{i,t}^{(q)}$ is the absolute value of $VR_{i,t}^{(q)}$ minus one. If $AbsVarRatio_{i,t}^{(q)}$ is equal to zero, it implies that the asset prices follow a random walk. On the other hand, if $AbsVarRatio_{i,t}^{(q)}$ has a non-negative value, it suggests evidence of price inefficiency. The magnitude of the non-negative value implies how large the price inefficiency is.

Previous works have followed this approach by modifying the research design to fit into their models. For example, Boehmer and Kelley (2009) use intraday returns and consider intraday horizons such as (5, 30), (5, 60), (10, 30), and (10, 60) minutes in a daily window to measure price inefficiency. For example, "(5, 30) minutes" indicates the VR value is the variance of 30-minute returns divided by six times the variance

of the 5-minute returns in a given day. They also consider longer horizons such as (1, 5), (1, 10), and (1, 20) days in the quarterly window. In addition, Ben-David et al. (2018) use a five-day return for the choice of q in the quarterly window. They estimate the absolute value of VR and regress it on their explanatory variable to examine the impact of ETF ownership on price efficiency. They find that ETF ownership leads to the deviation of stock prices from a random walk, implying the deterioration of stock price efficiency. Brogaard, Ringgenberg, and Sovich (2019) use a q-day overlapping horizon such as (1, 2), (1, 4), (1, 6), and (1, 8) days in the monthly window. They show their prediction that financialization worsens informativeness is consistent across different horizons. Overall, these methods allow them to show whether their main interest variables lead the asset prices to diverge from a random walk, implying price inefficiency.

Following these VR test empirical designs, we mainly focus on a one-day time window since the frequency of our main variables is at an intraday level. Specifically, in a one-day window, we use pairs of (5, 90), (5, 120), (10, 90), and (10, 120) minutes in the spirit of Boehmer and Kelley (2009). We use q-period overlapping horizon and adopt the absolute value of VR minus one as our dependent variable ($AbsVarRatio_{i,t}^{(q)}$). We then regress $AbsVarRatio_{i,t}^{(q)}$ on 0DTE%.

The estimated equation is:

$$AbsVarRatio_{i,t}^{(q)} = \alpha + \beta 0\widehat{DTE\%}_{\iota,t} + \delta' X_{i,t} + \omega_i + \gamma_t + \varepsilon_{i,t}, \quad \text{(IA.8)}$$

where $AbsVarRatio_{i,t}^{(q)}$ is defined as the absolute value of $VR_{i,t}^{(q)}$ minus one in a daily window using a q-periods overlapping horizon, and $0\widehat{DTE\%}_{\iota,t}$ is the estimated 0DTE% from the first-stage regression in equation (4). $X_{i,t}$ is a vector of control variables, ω_i is an index fixed effect, and γ_t is a day of the week fixed effect. The standard errors are clustered at the daily level.

The VR is adjusted by using an unbiased and efficient estimator of each variance following Lo and MacKinlay (1988). Lo and MacKinaly (1988) devise this estimator to overcome the limited sample size

and to improve the power of the VR test. Both dependent and independent variables are standardized. We expect that 0DTE% is positively associated with $AbsVarRatio_{i,t}^{(q)}$. This is because if retail investors are more active in 0DTE options trading (CBOE 2025, Bryzgalova et al. 2023, NYSE 2023), their speculative and uninformed trading behavior (Lakonishok et al. 2007; Liu et al. 2020; Beckmeyer et al. 2023) could introduce greater noise into the market and disrupt the incorporation of new information (Chiras and Manaster 1978; Bloomfield, O'Hara, and Saar 2009). Based on these previous empirical findings and theoretical models, increased 0DTE options trading may result in lower price efficiency (Stein 1987; Froot et al. 1992).

Table IA4 presents the results of the IV regressions.[1] Each column shows the results with a different pair of the VR test: (5, 90), (5, 120), (10, 90), and (10, 120) minutes. Across all specifications, the estimated coefficients of 0DTE% are positively associated with $AbsVarRatio_{i,t}^{(q)}$.[2] Columns (1) and (2), which use the higher frequency ratios of (5, 90) and (5, 120) minutes, show that the estimated coefficients are 0.061 and 0.084, statistically significant at the 10% and 5% levels, respectively. Similarly, columns (3) and (4), which use relatively lower frequency ratios of (10, 90) and (10, 120) minutes, report the estimated coefficients of 0.072 and 0.090, which are statistically significant at the 5% and 1% levels, respectively. If the 0DTE options trading contributes to informational efficiency, then the absolute value of VR would not be different from zero. However, since it is observed that the estimated coefficients are larger than zero in every specification, we find that an increase in 0DTE options trading drives prices further away from the random-walk. In sum, our results consistently show that $AbsVarRatio_{i,t}^{(q)}$ increases with 0DTE%, suggesting that 0DTE options trading leads to reduced price efficiency of the market.

[Insert Table IA4 here]

[1] Our results are robust when we use OLS regression instead of IV regression.

[2] Our results are robust when we use different time horizons, such as q = 20, 30, and 60 minutes.

IA2.2. 0DTE versus Non-0DTE

We have shown that 0DTE options trading increases volatility. Since 0DTE options trading impacts volatility, other short-term date expiration options, such as 1DTE or 2DTE options, may also have a similar impact. However, we hypothesize that 0DTE options exhibit the strongest explanation power for volatility, not only because of the large volume of retail investors in 0DTE options (CBOE 2025) but also because of the unique characteristics of 0DTE options that retail investors prefer, such as no overnight risk and the expiration occurring on the trading day (Boes et al. 2007; Bauer et al. 2009; Kelly and Clark 2011; Lou et al. 2019). These characteristics are proper for speculative investment because they allow investors to realize profit or loss quickly. Accordingly, we anticipate that retail investors, who are characterized by their noise trading and speculative behaviors (Lakonishok et al. 2007; Liu et al. 2020), prefer 0DTE to other short-term expiration options, so 0DTE options trading has a stronger impact on volatility than other short-term options trading.

As baseline results, we use equation (3) to examine the standalone impact of each short-term expiration option on volatility. We consider the short-term expiration options ranging from 0DTE to 4DTE, which account for one cycle of five business days in a given week. Columns (1) to (5) of Table IA5 display the results for each *n*DTE, where n = 0, 1, …, 4. The coefficient of 0DTE in column (1) is 0.056 and exhibits a positive relationship with volatility at the 1% level. Although the estimated coefficients of 1DTE and 2DTE show a positive and significant relationship with volatility, where the value is 0.043 and 0.015 in columns (2) and (3), respectively, the coefficient of 0DTE% displays the largest magnitude among others. The estimated coefficients for 3DTE and 4DTE in columns (4) and (5) show an insignificant relation with the volatility.

This result provides preliminary evidence that volatility is more responsive to 0DTE options compared to other short-term expiration options. However, to fairly compare the impact of short-term expiration options, we should include all *n*DTE options in a single regression. A key challenge to conducting this test is that multiple short-term expiration options, such as 0DTE, 1DTE, and 2DTE, are typically traded simultaneously, which may lead to multicollinearity bias.

To address this concern, we should extract the pure variation of each *n*DTE, which cannot be explained by other short-term expiration options. Accordingly, we regress each short-term expiration option on the others and obtain the estimated residuals. This is because residuals here can represent the unique variation of each *n*DTE option, net of the influence from other short-term expiration options. For instance, to capture the unique variation of 0DTE options, which is not explained by other *n*DTE options, we regress 0DTE on 1DTE through 4DTE and obtain the residual, $\epsilon_{0DTE,i,t}$.

The estimated equation is:

$$nDTE\%_{i,t} = \Sigma_{n \neq k} \beta_k kDTE\%_{i,t} + \delta' X_{i,t} + \omega_i + \gamma_t + \epsilon_{nDTE,i,t}, \quad \text{(IA.9)}$$

where $nDTE\%_{i,t}$ denotes the ratio of the trading volume of *n*DTE options over the total trading volume of the index-linked options and n, k = 0, 1, …, 4. $X_{i,t}$ is a vector of control variables, ω_i is an index fixed effect, and γ_t is a day of the week fixed effect. The standard errors are clustered at the daily level.

By using this equation, we calculate each residual, such as $\epsilon_{0DTE,i,t}$, $\epsilon_{1DTE,i,t}$, $\epsilon_{2DTE,i,t}$, $\epsilon_{3DTE,i,t}$, and $\epsilon_{4DTE,i,t}$. Then, we regress volatility on all residuals together with control variables to determine which short-term expiration option explains volatility the most strongly in the horse-race style approach.

The estimated equation is:

$$\sigma_{i,t} = \alpha + \Sigma_{n=0}^{4}\beta_n \epsilon_{nDTE,i,t} + \delta' X_{i,t} + \omega_i + \gamma_t + \xi_{i,t}, \quad \text{(IA.10)}$$

where $\sigma_{i,t}$ is the daily standard deviation of the index's midpoint quote return every 5 minutes, and $\epsilon_{nDTE,i,t}$ is the residual from equation (IA.9). $X_{i,t}$ is a vector of control variables, ω_i is an index fixed effect, and γ_t is a day of the week fixed effect. The standard errors are clustered at the daily level.

We provide the result of the horse-race test with all pure variations at the same time in column (6) of Table IA5.[3] The result documents that the pure variation of 0DTE ($\epsilon_{0DTE,i,t}$), which is not explained by other short-term expiration options, has the largest impact on the volatility, among others.[4] The coefficient of the pure variation of 0DTE is 0.056, surpassing those of other short-term expiration options. Specifically, the coefficients of the pure variation of 1DTE and 2DTE are 0.047 and 0.018. Additionally, the coefficient of the pure variation of 3DTE and 4DTE has a statistically insignificant relation with the volatility. These findings confirm that the impact on the volatility is most strongly concentrated in 0DTE options trading, suggesting that the nature of 0DTE options plays a unique and dominant role in driving market volatility relative to other short-term date expiration options.

[Insert Table IA5 here]

IA2.3. Alternative Volatility Measures

Next, we assess the robustness of our findings by employing alternative volatility measures to ensure our results are not sensitive to the specific 5-minute frequency used in the baseline analysis.

[3] To mitigate heteroskedasticity in the volatility and 0DTE options trading at different times, we provide results with alternative clustered standard errors in Panel C of Table IA9. The results are robust across alternative clustering choices.

[4] The results remain robust when we replace $\epsilon_{nDTE,i,t}$ with $nDTE\%_{i,t}$. We document that volatility is more responsive to 0DTE than other short-term expiration options, consistent with the results in column (6).

First, we replicate our analysis using volatility calculated at 10- and 30-minute intervals. In Table IA6, we find that 0DTE options trading consistently exhibits a positive and statistically significant impact on volatility across these different frequencies. Specifically, a one standard deviation increase in 0DTE% is estimated to increase the 10- and 30-minute volatilities by 13.29% and 20.66% relative to their mean values, respectively. It should be noted that the coefficients reported in Table IA6 are naturally larger than our main results; this is primarily attributed to the use of lower-frequency volatility measures, which result in different scaling compared to the 5-minute measure.

[Insert Table IA6 here]

Second, to address the concern that shorter-window realized volatility measures may be subject to microstructure noise, we compute a five-minute realized variance using overlapping one-minute returns, following Aït-Sahalia et al. (2005) and Bollerslev et al. (2022). This procedure smooths out idiosyncratic price movements and improves the precision of high-frequency volatility estimation. We document that the estimated coefficients of 0DTE options trading on this smoothed volatility measure remain positive, stable in magnitude, and statistically significant across all specifications. These findings reinforce that our main conclusions are not driven by microstructure noise or measurement error in short-interval volatility calculations. In Table IA7, we document that the estimated coefficients of 0DTE options trading on this volatility measure continue to remain positive, stable in magnitude, and statistically significant across all specifications. These findings reinforce that our main findings are not driven by microstructure noise or measurement error in short-interval volatility calculations.

[Insert Table IA7 here]

IA2.4. Alternative Econometric Specifications

In this section, we address the possibility that time-specific shocks are not adequately captured by the standard set of control variables. To do this, we implement a more granular specification that includes daily fixed effects. This approach helps control unobserved heterogeneity at the daily level. As reported in Table IA8, the results continue to hold even with this stringent specification. This further reduces the likelihood that our findings are driven by autocorrelation in volatility or specific daily trends. Overall, these tests provide strong evidence that our results are not mechanically driven by persistence in 0DTE% or volatility.

[Insert Table IA8 here]

In addition to alternative fixed effects, we examine the sensitivity of our statistical inference to the choice of standard error clustering. In our baseline specifications, we cluster standard errors at the daily level to account for potential serial correlation and heteroskedasticity within days. However, given the time-series nature of our data, there is a potential concern that correlation in residuals might persist over longer horizons (e.g., monthly), which could bias standard errors downward and inflate *t*-statistics. To mitigate this concern, we re-estimate our main results using alternative clustering dimensions: monthly, year-day, and month-day. Table IA9 reports the results across three different panels corresponding to our key findings: the second stage of the IV regression in Table 5 (Panel A), Table 6 (Panel B), and Table IA5 (Panel C), respectively. Across all specifications and panels, we find that the statistical significance of the 0DTE trading coefficients remains robust. Even when clustering at the monthly level, which imposes a more conservative penalty for serial correlation, the coefficients remain statistically significant at the 1% level. These results confirm that our inference is valid and that the significance of our findings is not an artifact of the standard error clustering method.

[Insert Table IA9 here]

IA2.5. Alternative Control Variables

To further address potential endogeneity concerns arising from simultaneity bias, we examine whether our OLS results are sensitive to the timing of the control variables. In our baseline specifications, control variables are measured contemporaneously. However, it is possible that current market conditions (i.e., trading volume or bid-ask spreads) and volatility are jointly determined within the same day. To mitigate this concern, we re-estimate our baseline OLS regression using one-day lagged control variables. Table IA10 presents the results of this specification. We find that the association between 0DTE option trading and market volatility remains positive and statistically significant at the 1% level across all columns, even when conditioning on the previous day's market characteristics. This suggests that the relationship is not driven by the simultaneous determination of control variables and volatility.

[Insert Table IA10 here]

We extend the robustness check regarding lagged controls to our IV framework. This ensures that the causal interpretation of our results is not confounded by the endogenous relationship between contemporaneous control variables and the introduction of daily expirations. By using lagged controls, we isolate the effect of the exogenous variation in 0DTE options trading driven by the listing of new expirations, independent of current market microstructure shifts. Table IA11 reports both the first-stage and second-stage results. Consistent with our main IV findings, the instrument remains strong, and the second-stage estimates for the 0DTE% coefficient remain positive and statistically significant at the 1% level. These results reinforce the

conclusion that the impact of 0DTE trading on volatility is robust to the timing of control variable measurement.

[Insert Table IA11 here]

While our baseline models control for one-day lagged volatility, there is a concern that a single lag may not fully capture the complex autocorrelation structure of volatility, potentially leading to omitted variable bias where 0DTE options trading for longer-term volatility trends. To address this, we estimate both OLS and IV specifications that include up to five days of lagged volatility. As shown in Table IA12, the coefficient on 0DTE% remains positive and statistically significant in both the OLS and IV specifications, even after absorbing significant variation from multiple past volatility terms. This provides strong evidence that the observed effect of 0DTE options trading is distinct from the natural persistence of market volatility.

[Insert Table IA12 here]

Internet Appendix Tables and Figures

Table IA1: Variable Definitions

Symbol	Definition
	The main independent and dependent variables
$\sigma_{i,t}$	The daily standard deviation of the index i's midpoint quote returns every 5 minutes between 9:30 AM and 4:00 PM Eastern Time, where $i \in$ {S&P 500, NASDAQ-100, Russell 2000}.
$\sigma_{i,t}^{10min}$	The daily standard deviation of the index i's midpoint quote returns every 10 minutes between 9:30 AM and 4:00 PM Eastern Time, where $i \in$ {S&P 500, NASDAQ-100, Russell 2000}.
$\sigma_{i,t}^{30min}$	The daily standard deviation of the index i's midpoint quote returns every 30 minutes between 9:30 AM and 4:00 PM Eastern Time, where $i \in$ {S&P 500, NASDAQ-100, Russell 2000}.
$nDTE\%_{i,t}$	The ratio of the total volume of nDTE (where $n = 1, 2, 3, and\ 4$) options traded over the total volume of the linked options for index i traded in each day expressed as a percentage, where $i \in$ {S&P 500, NASDAQ-100, Russell 2000}.
$Introduction_{i,t}$	The dummy variable equal to zero before the introduction of index i's weekly option and one afterward, where $i \in$ {S&P 500, NASDAQ-100, Russell 2000}. Details of the introduction date for each index are stated in Table 1.
$Outage_t$	The dummy variable equal to one if Robinhood has an outage during trading hours and zero otherwise.
$AbsVarRatio_{i,t}^{(q)}$	The absolute value of Variance Ratio minus 1 in time window T, using q-periods overlapping horizon. The Variance Ratio is defined as $VR_{i,t}^{(q)}$, which is the ratio of the variance of q-period returns divided by the variance of one-period return in the time window T. We adjust the VR by using an unbiased and efficient estimator of each variance following Lo and MacKinlay (1988). $$AbsVarRatio_{i,t}^{(q)} = \left\lvert VR_{i,t}^{(q)} - 1 \right\rvert, \quad VR_{i,t}^{(q)} = \frac{Var(r_{i,t-q\to t})}{Var(r_{i,t}) * q}$$
$\sigma_{i,t}^{HB}$	The daily standard deviation of the pricing error using second-by-second data from the TAQ following Hasbrouck (1993).
$\epsilon_{nDTE,i,t}$	The residual of regression of $nDTE\%_{i,t}$ on $\sum_{k\neq n} kDTE\%_{i,t}$, where n, k = 0, 1, …, 4. For example, $\epsilon_{0DTE,t}$ is the residual of the following regression: $0DTE\%_{i,t} = \alpha_0 + \alpha_1 1DTE\%_{i,t} + \alpha_2 2DTE\%_{i,t} + \alpha_3 3DTE\%_{i,t} + \alpha_4 4DTE\%_{i,t} + \epsilon_{0DTE,i,t}$ Using fitted value, $\epsilon_{0DTE,i,t} = 0DTE\%_{i,t} - \left(\hat{\alpha}_1 1DTE\%_{i,t} + \hat{\alpha}_2 2DTE\%_{i,t} + \hat{\alpha}_3 3DTE\%_{i,t} + \hat{\alpha}_4 4DTE\%_{i,t}\right)$
$\lvert Ret_{i,t}^{CC}\rvert$ or $\left(Ret_{i,t}^{CC}\right)^2$	The absolute or squared price appreciation between the market close of the day $t-1$ to the close of the same day t, following Lou et al. (2019).
$\lvert Ret_{i,t}^{OC}\rvert$ or $\left(Ret_{i,t}^{OC}\right)^2$	The absolute or squared price appreciation between market open and close of the same day t, following Lou et al. (2019).
$\lvert Ret_{i,t}^{Overnight}\rvert$ or $\left(Ret_{i,t}^{Overnight}\right)^2$	The absolute or squared non-trading-hours price change between the market close of day t and the market open of day $t+1$, constructed as the residual between close-to-close and intraday returns following Lou et al. (2019).
$\sigma_{i,t}^{1\,\min\, overlap}$	The daily standard deviation of the index's midpoint quote returns every 5 minutes using overlapping 1-minute returns following Aït-Sahalia et al. (2005) and Bollerslev et al. (2022).

Table IA1 Continued.

	Control Variables: Index level
$Index\ Return_{i,t}$	The daily return of the index i, where $i \in$ {S&P 500, NASDAQ-100, Russell 2000}.
$Index\ Volume_{i,t}$	The daily trading volume of the index i, where $i \in$ {S&P 500, NASDAQ-100, Russell 2000}. The unit is one million.
$\sigma_{i,t-n}$	The daily standard deviation of the index i's midpoint quote returns every 5 minutes between 9:30 AM and 4:00 PM Eastern Time n-days ago, where $i \in$ {S&P 500, NASDAQ-100, Russell 2000}.
	Control Variables: Macroeconomics level
$CP\ Spread_t$	The daily difference between the commercial paper (CP) and the 3-month Treasury bill rate.
$Default\ Spread_t$	The daily difference between the yield on BAA-rated corporate bonds and the yield on long-term US government bonds.
σ_t^{forex}	The standard deviation of the daily return of the foreign exchange rates over the previous 22-day rolling periods.
σ_t^{cpi}	The standard deviation of the previous 12-month returns based on the year-over-year returns of the Consumer Price Index (CPI).
EPU_t	The daily Economic Policy Uncertainty (EPU) index in Baker et al. (2016).

Table IA2: Discussion for Market Maker's Net Gamma

This paper examines whether our main IV results are driven by market makers' aggregate net gamma exposure. The table reports two-stage least squares regressions based on the baseline IV specification in equation (5), augmented with the net gamma variable or its orthogonalized components. Columns (1) and (2) show the baseline first and second stages when the net gamma exposure of option market makers, $\Gamma_{i,t}^{Net}$, is added as a control. Columns (3) and (4) use the orthogonalized 0DTE trading measure, $0DTE\%_{i,t}^{\perp\Gamma^{Net}}$, defined as the residual from equation (11), while columns (5) and (6) use the orthogonalized volatility measure, $\sigma_{i,t}^{\perp\Gamma^{Net}}$, defined as the residual from equation (12). Details of the control variables are stated in Appendix Table IA1. t-statistics are reported in parentheses. We report Kleibergen-Paap *F*-statistic following Kleibergen and Paap (2006). ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively. Standard errors are clustered at the daily level. The sample covers the period from January 2010 to August 2023.

	Controls $\Gamma_{i,t}^{Net}$		Orthogonalize $0DTE\%_{i,t}$		Orthogonalize $\sigma_{i,t}$	
	1st stage	2nd stage	1st stage	2nd stage	1st stage	2nd stage
Dependent variable=	$0DTE\%_{i,t}$	$\sigma_{i,t}$	$0DTE\%_{i,t}^{\perp\Gamma^{Net}}$	$\sigma_{i,t}$	$0DTE\%_{i,t}$	$\delta_{i,t}^{\perp\Gamma^{Net}}$
	(1)	(2)	(3)	(4)	(5)	(6)
Introduction	21.946***		21.946***		21.946***	
	(59.25)		(59.25)		(59.25)	
$0\widehat{DTE\%}_{i,t}$ ($0\widehat{DTE\%}_{i,t}^{\perp\Gamma^{Net}}$)		0.063***		0.063***		0.063***
		(11.87)		(11.62)		(11.88)
$\Gamma_{i,t}^{Net}$	-0.052	-0.980***				
	(-0.85)	(-18.89)				
$Index\ Return_{i,t}$	-0.092	-0.485***	0.115	-0.626***	-0.099	0.014
	(-1.11)	(-6.50)	(1.41)	(-8.46)	(-1.21)	(0.19)
$Index\ Volume_{i,t}$	-0.025***	0.046***	0.022***	0.045***	-0.024***	0.003*
	(-12.23)	(26.42)	(11.09)	(24.74)	(-12.15)	(1.67)
$\sigma_{i,t-5}$	0.156***	0.353***	-0.200***	0.419***	0.158***	-0.023
	(5.77)	(19.51)	(-7.54)	(23.75)	(5.96)	(-1.27)
$CP\ Spread_t$	-4.999***	3.626***	0.221	3.564***	-4.985***	0.329
	(-4.99)	(6.88)	(0.22)	(6.87)	(-4.98)	(0.62)
$Default\ Spread_t$	-3.501***	-0.667***	5.472***	-1.053***	-3.492***	0.566***
	(-14.98)	(-4.53)	(23.55)	(-7.04)	(-15.02)	(3.87)
EPU_t	0.993***	0.824***	-1.164***	0.948***	0.993***	-0.136
	(5.77)	(8.17)	(-6.76)	(9.12)	(5.77)	(-1.35)
σ_t^{forex}	4.833***	1.392**	-6.112***	1.557**	4.806***	-0.689
	(3.94)	(2.34)	(-4.99)	(2.52)	(3.93)	(-1.16)
σ_t^{cpi}	5.678***	-0.285*	-0.888**	0.365**	5.690***	-0.415***
	(15.87)	(-1.82)	(-2.50)	(2.41)	(15.99)	(-2.70)
Observations	9,047	9,047	9,047	9,047	9,047	9,047
Index FEs	Yes	Yes	Yes	Yes	Yes	Yes
Day-of-the-week FEs	Yes	Yes	Yes	Yes	Yes	Yes
Adjusted R-squared		0.609		0.581		0.028
Kleibergen-Paap F-statistic		3,510		3,511		3,511

Table IA3: Discussion for Volume and Order Imbalance

This table examines whether the impact of 0DTE options trading depends on order imbalance and option market makers' net gamma exposure. The regressions augment the baseline specifications in Table 3 (OLS) and Table 4 (second stage of the IV) by including the net gamma measure, $\Gamma_{i,t}^{Net}$, and its interactions with 0DTE options trading, $0DTE\%_{i,t}$, and 0DTE put–call ratio, $0DTE\ PCR_{i,t}$. Column (1) reports OLS estimates by estimating equation (3), while column (2) reports the corresponding second-stage IV estimates by estimating equation (5). Details of the control variables are stated in Appendix Table IA1. t-statistics are reported in parentheses. We report Kleibergen-Paap *F*-statistic following Kleibergen and Paap (2006). ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively. Standard errors are clustered at the daily level. The sample covers the period from January 2010 to August 2023.

	(1)	(2)
	Table 3 OLS	Table 4 2nd-stage
Dependent variable=	$\sigma_{i,t}$	$\sigma_{i,t}$
$0DTE\%_{i,t}$ ($\widehat{0DTE\%}_{i,t}$)	0.0751***	0.1758***
	(7.84)	(4.06)
$0DTE\ PCR_{i,t}$	3.7897**	4.6136**
	(2.11)	(2.46)
$\Gamma_{i,t}^{Net}$	0.4226	0.4903
	(0.43)	(0.51)
$0DTE\%_{i,t}$ ($\widehat{0DTE\%}_{i,t}$) $\times\ \Gamma_{i,t}^{Net}$	0.3229	0.9005
	(0.18)	(0.52)
$0DTE\ PCR_{i,t} \times \Gamma_{i,t}^{Net}$	-0.1433***	-0.1525***
	(-4.63)	(-4.93)
$Index\ Return_{i,t}$	-0.2905*	-0.2578*
	(-1.92)	(-1.68)
$Index\ Volume_{i,t}$	0.0647***	0.0711***
	(15.43)	(13.77)
$\sigma_{i,t-5}$	0.3244***	0.3043***
	(11.74)	(10.16)
$CP\ Spread_t$	2.1776***	3.0135***
	(2.88)	(3.72)
$Default\ Spread_t$	-1.2208***	-0.7057*
	(-3.67)	(-1.79)
EPU_t	0.7390***	0.4740**
	(4.52)	(2.46)
σ_t^{forex}	1.6815	0.1212
	(1.36)	(0.08)
σ_t^{cpi}	-0.5079**	-1.4693***
	(-2.01)	(-3.04)
Observations	2,322	2,322
Adjusted R-squared	0.6150	0.5874
Index FE	Yes	Yes
Day-of-the-week FE	Yes	Yes
Kleibergen-Paap F-statistic		101.8

Table IA4: Price Efficiency: Variance Ratio Test

The table reports the impact of 0DTE options trading on price efficiency using the Variance Ratio (VR) test. The VR is denoted as $VR_{i,t}^{(q)}$, which is the ratio of the variance of q-period returns divided by q times the variance of one-period return in time window T, by using an unbiased estimator following Lo and MacKinlay (1988). The regression is

$$AbsVarRatio_{i,t}^{(q)} = \alpha + \beta 0\widehat{DTE\%}_{i,t} + \delta' X_{i,t} + \omega_i + \gamma_t + \varepsilon_{i,t},$$

where $AbsVarRatio_{i,t}^{(q)}$ is defined as the absolute value of $VR_{i,t}^{(q)}$ minus 1 in a daily window using a q-periods overlapping horizon and $0\widehat{DTE\%}_{i,t}$ is the estimated 0DTE% from the first-stage regression in equation (4). $X_{i,t}$ is a vector of control variables, ω_i is an index fixed effect, and γ_t is a day of the week fixed effect. Details of the control variables are stated in Appendix Table IA1. Both dependent variables and independent variables are standardized. We report Kleibergen-Paap *F*-statistic following Kleibergen and Paap (2006). The *t*-statistics are presented in parentheses. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively. The standard errors are clustered at the daily level. The sample covers the period from January 2010 to August 2023.

Dependent variable =	$AbsVarRatio_{i,t}^{(q)}$			
(*t*, *q*) =	(5,90)	(5,120)	(10,90)	(10,120)
	(1)	(2)	(3)	(4)
$0\widehat{DTE\%}_{i,t}$	0.061*	0.084**	0.072**	0.090***
	(1.88)	(2.47)	(2.15)	(2.58)
$Index\ Return_{i,t}$	0.087***	0.092***	0.077***	0.086***
	(3.46)	(2.99)	(2.82)	(2.59)
$Index\ Volume_{i,t}$	0.075***	0.104***	0.071***	0.097***
	(4.16)	(5.17)	(3.81)	(4.66)
$\sigma_{i,t-5}$	-0.064***	-0.069***	-0.065***	-0.068***
	(-3.42)	(-3.55)	(-3.40)	(-3.44)
$CP\ Spread_t$	-0.025	-0.026	-0.034	-0.031
	(-0.85)	(-0.74)	(-1.06)	(-0.86)
$Default\ Spread_t$	-0.013	0.004	-0.001	0.013
	(-0.55)	(0.16)	(-0.05)	(0.55)
EPU_t	-0.026	-0.027	-0.022	-0.020
	(-1.49)	(-1.48)	(-1.18)	(-1.08)
σ_t^{forex}	-0.002	0.010	-0.002	0.009
	(-0.07)	(0.44)	(-0.11)	(0.36)
σ_t^{cpi}	0.002	-0.014	-0.002	-0.016
	(0.12)	(-0.65)	(-0.10)	(-0.73)
Observations	9,047	9,047	9,047	9,047
Adjusted R-squared	0.010	0.012	0.009	0.011
Index FE	Yes	Yes	Yes	Yes
Days of week FE	Yes	Yes	Yes	Yes
Kleibergen-Paap *F*-statistic	2,350	2,350	2,350	2,350

Table IA5: 0DTE vs Non-0DTE

The table presents a causal impact of 0DTE% to 4DTE% on volatility. 1DTE% to 4DTE% represents the non-0DTE%. In columns (1) to (5), we use equation (3) to measure the standalone impact of *n*DTE%. The regression is

$$\sigma_{i,t} = \alpha + \beta\, nDTE\%_{i,t} + \delta' X_{i,t} + \omega_i + \gamma_t + \varepsilon_{i,t},$$

where n = 0, 1, …, 4. Next, in column (6), to isolate the pure effect of each *n*DTE%, we regress each *n*DTE% on other short-term expiration options and obtain the residual, $\epsilon_{nDTE_{i,t}}$ is the estimated residual from equation (IA.9). Then, we run the following regression. The regression is

$$\sigma_{i,t} = \alpha + \Sigma_{n=0}^{4}\beta_n \epsilon_{nDTE,i,t} + \delta' X_{i,t} + \omega_i + \gamma_t + \xi_{i,t},,$$

where $\sigma_{i,t}$ is the daily standard deviation of the index's midpoint quote return every 5 minutes, $\epsilon_{nDTE,i,t}$ is the estimated residual from equation (IA.9), and n = 0, 1, …, 4. $X_{i,t}$ is a vector of control variables, ω_i is an index fixed effect, and γ_t is a day of the week fixed effect. Details of the control variables are stated in Appendix Table IA1. The *t*-statistics are presented in parentheses. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively. The standard errors are clustered at the daily level. The sample covers the period from January 2010 to August 2023.

Dependent variable =	$\sigma_{i,t}$					
	(1)	(2)	(3)	(4)	(5)	(6)
$0DTE\%_{i,t}$ $(\epsilon_{0DTE,i,t})$	0.056***					0.056***
	(11.15)					(11.40)
$1DTE\%_{i,t}$ $(\epsilon_{1DTE,i,t})$		0.043***				0.047***
		(7.24)				(7.42)
$2DTE\%_{i,t}$ $(\epsilon_{2DTE,i,t})$			0.015***			0.018***
			(2.63)			(2.76)
$3DTE\%_{i,t}$ $(\epsilon_{3DTE,i,t})$				0.007		0.008
				(1.15)		(1.20)
$4DTE\%_{i,t}$ $(\epsilon_{4DTE,i,t})$					-0.005	-0.005
					(-0.96)	(-1.01)
$Index\ Return_{i,t}$	-0.614***	-0.619***	-0.624***	-0.626***	-0.625***	-0.626***
	(-8.27)	(-8.20)	(-8.21)	(-8.24)	(-8.23)	(-8.54)
$Index\ Volume_{i,t}$	0.047***	0.046***	0.045***	0.045***	0.045***	0.045***
	(25.65)	(24.97)	(24.11)	(24.08)	(24.10)	(25.01)
$\sigma_{i,t-5}$	0.399***	0.411***	0.419***	0.419***	0.419***	0.419***
	(22.23)	(22.61)	(22.70)	(22.69)	(22.75)	(24.27)
$CP\ Spread_t$	3.854***	3.503***	3.578***	3.561***	3.565***	3.564***
	(7.37)	(6.44)	(6.51)	(6.46)	(6.46)	(7.04)
$Default\ Spread_t$	-0.555***	-0.851***	-1.032***	-1.049***	-1.056***	-1.053***
	(-3.61)	(-5.45)	(-6.67)	(-6.77)	(-6.83)	(-7.14)
EPU_t	0.828***	0.907***	0.944***	0.949***	0.946***	0.948***
	(7.99)	(8.63)	(8.79)	(8.83)	(8.79)	(9.41)
σ_t^{forex}	0.951	1.555**	1.573**	1.571**	1.556**	1.557**
	(1.54)	(2.43)	(2.44)	(2.44)	(2.41)	(2.56)
σ_t^{cpi}	-0.000	0.245	0.374**	0.372**	0.358**	0.365**
	(-0.00)	(1.60)	(2.45)	(2.44)	(2.34)	(2.48)
Observations	9,047	9,047	9,047	9,047	9,047	9,047
Adjusted R-squared	0.582	0.574	0.571	0.570	0.570	0.591
Index FE	Yes	Yes	Yes	Yes	Yes	Yes
Days of week FE	Yes	Yes	Yes	Yes	Yes	Yes

Table IA6: Alternative Volatility Intervals

The table presents a causal impact of 0DTE options trading and volatility, using alternative frequency of volatility measures. The regression is

$$Y_{i,t} = \alpha + \beta\ 0\widehat{DTE\%}_{i,t} + \delta' X_{i,t} + \omega_i + \gamma_t + \varepsilon_{i,t},$$

where $Y_{i,t}$ takes one of our main dependent variables: $\sigma_{i,t}^{10min}$ or $\sigma_{i,t}^{30min}$. $\sigma_{i,t}^{10min}$ and $\sigma_{i,t}^{30min}$ are the daily standard deviation of the index's midpoint quote return every 10 minutes and 30 minutes, respectively. $0\widehat{DTE\%}_{i,t}$ is the estimated 0DTE% from the first-stage regression in equation (4), $X_{i,t}$ is a vector of control variables, ω_i is an index fixed effect, and γ_t is a day of the week fixed effect. Details of the control variables are stated in Appendix Table IA1. The *t*-statistics are presented in parentheses. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively. We report Kleibergen-Paap *F*-statistic following Kleibergen and Paap (2006). The standard errors are clustered at the daily level. The sample covers the period from January 2010 to August 2023.

Dependent variable =	$\sigma_{i,t}^{10min}$	$\sigma_{i,t}^{30min}$
	(1)	(2)
$0\widehat{DTE\%}_{i,t}$	0.092***	0.143***
	(11.76)	(10.01)
$Index\ Return_{i,t}$	-0.830***	-1.218***
	(-7.98)	(-6.62)
$Index\ Volume_{i,t}$	0.067***	0.105***
	(25.58)	(23.32)
$\sigma_{i,t-5}$	0.551***	0.881***
	(20.99)	(18.46)
$CP\ Spread_t$	5.484***	8.290***
	(7.61)	(6.07)
$Default\ Spread_t$	-0.674***	-1.176***
	(-3.02)	(-2.99)
EPU_t	1.098***	1.649***
	(7.58)	(6.29)
σ_t^{forex}	1.486*	2.305
	(1.65)	(1.38)
σ_t^{cpi}	-0.071	0.135
	(-0.31)	(0.33)
Observations	9,047	9,047
Adjusted R-squared	0.564	0.477
Index FE	Yes	Yes
Days of week FE	Yes	Yes
Kleibergen-Paap *F*-statistic	3,511	3,511

Table IA7: Alternative Robustness for Different Measures of Volatility

The table suggests evidence about how the lagged control variables affect the main findings in Table 5. The table presents a causal impact of 0DTE options trading and volatility using IV regression. The first-stage regression is

$$0DTE\%_{i,t} = \alpha + \beta\ introduction_{i,t} + \delta' X_{i,t} + \omega_i + \gamma_t + u_{i,t},$$

where $0DTE\%_{i,t}$ is computed as the ratio of the total volume of 0DTE options traded over the total trading volume of the index-linked options expressed as a percentage, $introduction_{i,t}$ is the dummy set to zero before the introduction of the index weekly option and one afterward, $X_{i,t}$ is a vector of control variables, ω_i is an index fixed effect, and γ_t is a day of the week fixed effect. The second-stage regression is

$$\sigma_{i,t}^{1\,\min\,overlap} = \alpha + \beta\ \widehat{0DTE\%}_{i,t} + \delta' X_{i,t} + \omega_i + \gamma_t + \varepsilon_{i,t},$$

where $\sigma_{i,t}^{1\,\min\,overlap}$ is the daily standard deviation of the index's midpoint quote return every 5 minutes using overlapping 1-minute returns following Aït-Sahalia et al. (2005) and Bollerslev et al. (2022), $\widehat{0DTE\%}_{i,t}$ is the estimated 0DTE% from the first-stage regression. Columns (1) to (3) report the estimates from the first stage of IV regression, and columns (4) to (6) report the estimates from the second stage of IV regression, where the dependent variable is the $\sigma_{i,t}^{1\,\min\,overlap}$, with Kleibergen-Paap *F*-statistic following Kleibergen and Paap (2006). Details of the control variables are stated in Appendix Table IA1. The *t*-statistics are presented in parentheses. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively. The standard errors are clustered at the daily level. The sample covers the period from January 2010 to August 2023.

Dependent variable =	First-stage estimates $0DTE\%_{i,t}$			Second-stage estimates $\sigma_{i,t}^{1\,\min\,overlap}$		
	(1)	(2)	(3)	(4)	(5)	(6)
$Introduction_{i,t}$	27.271***	26.550***	20.728***			
	(71.77)	(67.92)	(62.99)			
$\widehat{0DTE\%}_{i,t}$		-0.074	-0.017	0.078***	0.070***	0.042***
		(-0.74)	(-0.22)	(11.55)	(12.46)	(7.28)
$Index\ Return_{i,t}$		-0.019***	-0.012***			
		-0.074	-0.017		-0.724***	-0.742***
		(-0.74)	(-0.22)		(-4.67)	(-4.90)
$Index\ Volume_{i,t}$		-0.019***	-0.012***		0.047***	0.049***
		(-8.74)	(-5.83)		(19.34)	(19.37)
$\sigma_{i,t-5}^{1\,\min\,overlap}$		0.186***	0.044**		0.499***	0.437***
		(6.11)	(2.04)		(14.23)	(12.61)
$CP\ Spread_t$			-1.658***			-0.294*
			(-6.74)			(-1.73)
$Default\ Spread_t$			-2.677***			-0.420***
			(-23.76)			(-6.36)
EPU_t			0.514***			0.820***
			(3.41)			(7.65)
σ_t^{forex}			3.059***			2.948***
			(2.93)			(4.64)
σ_t^{cpi}			7.005***			-0.374**
			(24.36)			(-2.22)
Observations	9,635	9,580	9,432	9,635	9,580	9,432
Adjusted R-squared	Yes	Yes	Yes	0.024	0.553	0.576
Index FE	Yes	Yes	Yes	Yes	Yes	Yes
Days of week FE	Daily	Daily	Daily	Yes	Yes	Yes
Kleibergen-Paap *F*-statistic	5,151	4,613	3,967			

Table IA8: Alternative Robustness for Fixed Effects

The table suggests evidence about how the fixed effects affect the main findings in Table 5. Columns (1) to (3) report the estimates from the first stage of IV regression, and columns (4) to (6) report the estimates from the second stage of IV regression, where the dependent variable is the σ_{it}, with Kleibergen-Paap *F*-statistic following Kleibergen and Paap (2006). Details of the control variables are stated in Appendix Table IA1. The *t*-statistics are presented in parentheses. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively. The standard errors are clustered at the daily level. The sample covers the period from January 2010 to August 2023.

Dependent variable =	First-stage estimates $0DTE\%_{i,t}$			Second-stage estimates $\sigma_{i,t}$		
	(1)	(2)	(3)	(4)	(5)	(6)
$Introduction_{i,t}$	21.946***	17.973***	17.194***			
	(59.25)	(54.93)	(56.88)			
$\widehat{0DTE\%}_{i,t}$				0.063***	0.015***	0.007**
				(11.62)	(2.90)	(2.23)
$Index\ Return_{i,t}$	-0.099	-0.023	0.147	-0.613***	-0.497***	-0.337***
	(-1.21)	(-0.35)	(0.77)	(-8.27)	(-8.97)	(-5.70)
$Index\ Volume_{i,t}$	-0.024***	-0.005**	-0.004**	0.047***	0.042***	0.005***
	(-12.15)	(-2.47)	(-2.05)	(25.66)	(25.98)	(7.40)
$\sigma_{i,t-5}$	0.158***	-0.048*	-0.221***	0.397***	0.063***	0.532***
	(5.96)	(-1.71)	(-4.49)	(22.05)	(3.69)	(35.22)
$CP\ Spread_t$	-4.985***	-0.622		3.892***	-0.005	
	(-4.98)	(-0.50)		(7.49)	(-0.01)	
$Default\ Spread_t$	-3.492***	0.058		-0.490***	6.394***	
	(-15.02)	(0.04)		(-3.16)	(6.73)	
EPU_t	0.993***	0.011		0.812***	0.302***	
	(5.77)	(0.05)		(7.87)	(2.60)	
σ_t^{forex}	4.806***	-0.996		0.871	-0.700	
	(3.93)	(-0.45)		(1.41)	(-0.68)	
σ_t^{cpi}	5.690***			-0.049		
	(15.99)			(-0.31)		
Observations	9,047	9,047	10,005	9,047	9,047	10,005
Adjusted R-squared				0.581	0.282	0.313
Index FE	Yes	Yes	Yes	Yes	Yes	Yes
Days of week FE	Yes	Yes	No	Yes	Yes	No
Monthly FE	No	Yes	No	No	Yes	No
Daily FE	No	No	Yes	No	No	Yes
Kleibergen-Paap *F*-statistic	3,511	3,018	3,235			

Table IA9: Alternative Robustness for Clustered Standard Errors

The table suggests evidence of how the clustering way of standard errors affects the main findings in Tables 5, 6, and IA5 with different control variable settings. We report the estimates from the second stage of IV regression from Table 5. The *t*-statistics are presented in parentheses. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively. The sample covers the period from January 2010 to August 2023.

Panel A: Table 5				
	(1)	(2)	(3)	(4)
$0DTE\%_{i,t}$	0.063***	0.063***	0.063***	0.063***
	(11.62)	(7.88)	(10.00)	(12.46)
Observations	9,047	9,047	9,047	9,047
Adjusted R-squared	0.581	0.581	0.581	0.581
Index FE	Yes	Yes	Yes	Yes
Days of week FE	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes
Clustered Standard Errors	Daily	Monthly	Year-day	Month-day
Panel B: Table 6				
	(1)	(2)	(3)	(4)
$0\widehat{DTE\%}_{i,t}$	0.241***	0.241***	0.241***	0.241***
	(9.07)	(7.68)	(7.94)	(9.45)
Observations	8,803	8,803	8,803	8,803
Adjusted R-squared	0.181	0.181	0.181	0.181
Index FE	Yes	Yes	Yes	Yes
Days of week FE	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes
Clustered Standard Errors	Daily	Monthly	Year-day	Month-day
Panel C: Table IA5				
	(1)	(2)	(3)	(4)
$\epsilon_{0DTE,i,t}$	0.056***	0.056***	0.056***	0.056***
	(11.40)	(7.24)	(11.13)	(11.75)
$\epsilon_{1DTE,i,t}$	0.047***	0.047***	0.047***	0.047***
	(7.42)	(7.48)	(7.07)	(7.75)
$\epsilon_{2DTE,i,t}$	0.018***	0.018***	0.018***	0.018***
	(2.76)	(2.91)	(2.70)	(2.97)
$\epsilon_{3DTE,i,t}$	0.008	0.008	0.008	0.008
	(1.20)	(1.39)	(1.16)	(1.27)
$\epsilon_{4DTE,i,t}$	-0.005	-0.005	-0.005	-0.005
	(-1.01)	(-1.04)	(-1.11)	(-1.03)
Observations	9,047	9,047	9,047	9,047
Adjusted R-squared	0.591	0.591	0.591	0.591
Index FE	Yes	Yes	Yes	Yes
Days of week FE	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes
Clustered Standard Errors	Daily	Monthly	Year-day	Month-day

Table IA10: Alternative Robustness for Lagged Controls in OLS

The table suggests evidence about how the lagged control variables affect the main findings in Table 4. The table presents the association between 0DTE options trading and volatility. The regression is

$$\sigma_{i,t} = \alpha + \beta\ 0DTE\%_{i,t} + \delta' X_{i,t-1} + \omega_i + \gamma_t + \varepsilon_{i,t},$$

where $\sigma_{i,t}$ is the daily standard deviation of the index's midpoint quote return every 5 minutes, $0DTE\%_{i,t}$ is computed as the ratio of the total volume of 0DTE options traded over the total trading volume of the index-linked options expressed as a percentage, $X_{i,t-1}$ is a vector of one-day lagged control variables, ω_i is an index fixed effect, and γ_t is a day of the week fixed effect. Details of the control variables are stated in Appendix Table IA1. The *t*-statistics are presented in parentheses. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively. The standard errors are clustered at the daily level. The sample covers the period from January 2010 to August 2023.

Dependent variable =	$\sigma_{i,t}$		
	(1)	(2)	(3)
$0DTE\%_{i,t}$	0.068***	0.055***	0.029***
	(12.54)	(12.44)	(5.91)
$Index\ Return_{i,t-1}$		-0.717***	-0.738***
		(-7.70)	(-8.41)
$Index\ Volume_{i,t-1}$		0.037***	0.037***
		(20.47)	(18.92)
$\sigma_{i,t-6}$		0.484***	0.420***
		(25.94)	(22.91)
$CP\ Spread_{t-1}$			-0.201
			(-1.19)
$Default\ Spread_{t-1}$			-0.367***
			(-5.51)
EPU_{t-1}			0.744***
			(7.65)
σ_{t-1}^{forex}			2.607***
			(4.01)
σ_{t-1}^{cpi}			0.142
			(0.89)
Observations	10,010	10,002	9,850
Adjusted R-squared	0.027	0.488	0.508
Index FE	Yes	Yes	Yes
Days of week FE	Yes	Yes	Yes

Table IA11: Alternative Robustness for Lagged Controls in IV

The table suggests evidence about how the lagged control variables affect the main findings in Table 5. The table presents a causal impact of 0DTE options trading and volatility using IV regression. The first-stage regression is

$$0DTE\%_{i,t} = \alpha + \beta\ introduction_{i,t} + \delta' X_{i,t-1} + \omega_i + \gamma_t + u_{i,t},$$

where $0DTE\%_{i,t}$ is computed as the ratio of the total volume of 0DTE options traded over the total trading volume of the index-linked options expressed as a percentage, $introduction_{i,t}$ is the dummy set to zero before the introduction of the index weekly option and one afterward, $X_{i,t-1}$ is a vector of one-day lagged control variables, ω_i is an index fixed effect, and γ_t is a day of the week fixed effect. The second-stage regression is

$$\sigma_{i,t} = \alpha + \beta\ 0\widehat{DTE\%}_{\iota,t} + \delta' X_{i,t-1} + \omega_i + \gamma_t + \varepsilon_{i,t},$$

where $\sigma_{i,t}$ is the daily standard deviation of the index's midpoint quote return every 5 minutes, $0\widehat{DTE\%}_{\iota,t}$ is the estimated 0DTE% from the first-stage regression, $X_{i,t-1}$ is a vector of one-day lagged control variables, ω_i is an index fixed effect, and γ_t is a day of the week fixed effect. Columns (1) to (3) report the estimates from the first stage of IV regression, and columns (4) to (6) report the estimates from the second stage of IV regression, where the dependent variable is the σ_{it}, with Kleibergen-Paap *F*-statistic following Kleibergen and Paap (2006). The *t*-statistics are presented in parentheses. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively. The standard errors are clustered at the daily level. The sample covers the period from January 2010 to August 2023.

Dependent variable =	First-stage estimates $0DTE\%_{i,t}$			Second-stage estimates $\sigma_{i,t}$		
	(1)	(2)	(3)	(4)	(5)	(6)
$Introduction_{i,t}$	27.268***	26.343***	20.583***			
	(71.92)	(67.52)	(63.51)			
$0\widehat{DTE\%}_{\iota,t}$				0.069***	0.060***	0.036***
				(11.42)	(12.58)	(6.25)
$Index\ Return_{i,t-1}$		-0.140	-0.071		-0.715***	-0.737***
		(-1.46)	(-0.88)		(-7.66)	(-8.39)
$Index\ Volume_{i,t-1}$		-0.022***	-0.014***		0.037***	0.037***
		(-9.80)	(-7.01)		(20.34)	(18.91)
$\sigma_{i,t-6}$		0.231***	0.064***		0.481***	0.419***
		(9.10)	(2.78)		(26.03)	(22.96)
$CP\ Spread_{t-1}$			-1.662***			-0.171
			(-6.98)			(-1.01)
$Default\ Spread_{t-1}$			-2.664***			-0.338***
			(-24.54)			(-4.96)
EPU_{t-1}			0.462***			0.738***
			(3.16)			(7.58)
σ_{t-1}^{forex}			2.785***			2.565***
			(2.73)			(3.94)
σ_{t-1}^{cpi}			6.921***			0.090
			(24.44)			(0.56)
Observations	10,010	10,002	9,850	10,010	10,002	9,850
Adjusted R-squared				0.027	0.488	0.507
Index FE	Yes	Yes	Yes	Yes	Yes	Yes
Days of week FE	Yes	Yes	Yes	Yes	Yes	Yes
Kleibergen-Paap *F*-statistic	5,172	4,559	4,033			

Table IA12: Alternative Robustness for Multiple Lagged Volatility Controls

The table suggests evidence about how the inclusion of multiple lagged volatilities affects the main findings in Tables 4 and 5, respectively. Columns (1) report the estimates from the OLS regression (i.e., Table 4), and columns (2) report the estimates from the second stage of IV regression (i.e., Table 5), where the dependent variable is the σ_{it}. Details of the control variables are stated in Appendix Table IA1. The *t*-statistics are presented in parentheses. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively. The standard errors are clustered at the daily level. The sample covers the period from January 2010 to August 2023.

	(1)	(2)
Specification=	Table 4 OLS	Table 5 IV 2nd-stage
Dependent variable=	$\sigma_{i,t}$	$\sigma_{i,t}$
$0DTE\%_{i,t}$ ($\widehat{0DTE\%}_{i,t}$)	0.015***	0.010**
	(4.21)	(2.45)
$\sigma_{i,t-1}$	0.453***	0.454***
	(21.12)	(21.12)
$\sigma_{i,t-2}$	0.183***	0.183***
	(8.48)	(8.46)
$\sigma_{i,t-3}$	0.024	0.024
	(1.08)	(1.08)
$\sigma_{i,t-4}$	0.045**	0.045**
	(2.18)	(2.19)
$\sigma_{i,t-5}$	0.062***	0.062***
	(3.47)	(3.48)
$Index\ Return_{i,t}$	-0.753***	-0.753***
	(-14.55)	(-14.55)
$Index\ Volume_{i,t}$	0.027***	0.027***
	(18.98)	(18.86)
$CP\ Spread_t$	-0.279**	-0.305***
	(-2.40)	(-2.61)
$Default\ Spread_t$	-0.244***	-0.270***
	(-5.68)	(-6.00)
EPU_t	0.290***	0.295***
	(4.00)	(4.06)
σ_t^{forex}	0.590	0.623
	(1.25)	(1.31)
σ_t^{cpi}	-0.226**	-0.182*
	(-2.09)	(-1.70)
Observations	9,047	9,047
Adjusted R-squared	0.608	0.576
Index FE	Yes	Yes
Days of week FE	Yes	Yes